JavaScript Array Reference

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Complete Array Reference

Revised July 2025

Name	Description
[]	Creates a new Array
new Array()	Creates a new Array
at()	Returns an indexed element of an array
concat()	Joins arrays and returns an array with the joined arrays
constructor	Returns the function that created the Array prototype
copyWithin()	Copies array elements within the array, to and from specified positions
entries()	Returns a key/value pair Array Iteration Object
every()	Checks if every element in an array pass a test
fill()	Fill the elements in an array with a static value

filter()	Creates a new array with every element in an array that pass a test
find()	Returns the value of the first element in an array that pass a test
findIndex()	Returns the index of the first element in an array that pass a test
findLast()	Returns the value of the last element in an array that pass a test
findLastIndex()	Returns the index of the last element in an array that pass a test
flat()	Concatenates sub-array elements
flatMap()	Maps all array elements and creates a new flat array
forEach()	Calls a function for each array element
from()	Creates an array from an object
includes()	Check if an array contains the specified element
indexOf()	Search the array for an element and returns its position
isArray()	Checks whether an object is an array
join()	Joins all elements of an array into a string
keys()	Returns a Array Iteration Object, containing the keys of the original array
lastIndexOf()	Search the array for an element, starting at the end, and returns its position
length	Sets or returns the number of elements in an array
map()	Creates a new array with the result of calling a function for each array element
of()	Creates an array from a number of arguments
pop()	Removes the last element of an array, and returns that element
prototype	Allows you to add properties and methods to an Array object
push()	Adds new elements to the end of an array, and returns the new length
reduce()	Reduce the values of an array to a single value (going left-to-right)
reduceRight()	Reduce the values of an array to a single value (going right-to-left)
reverse()	Reverses the order of the elements in an array
shift()	Removes the first element of an array, and returns that element
slice()	Selects a part of an array, and returns the new array
some()	Checks if any of the elements in an array pass a test
sort()	Sorts the elements of an array
splice()	Adds or Removes array elements

toReversed()	Reverses the order of array elements (to a new array)
toSorted()	Sorts the elements of an array (to a new array)
toSpliced()	Adds or Removes array elements (to a new array)
toString()	Converts an array to a string, and returns the result
unshift()	Adds new elements to the beginning of an array, and returns the new length
valueOf()	Returns the primitive value of an array
with()	Returns a new array with updated elements

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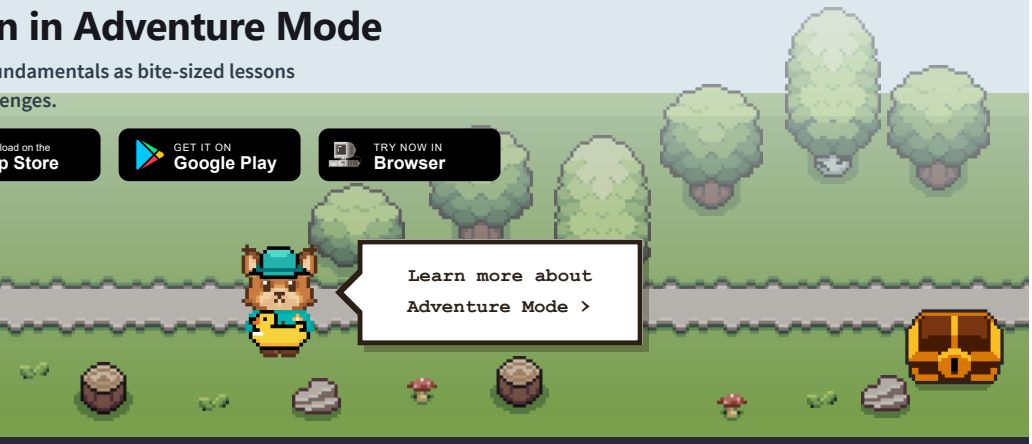
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JavaScript Operator Precedence

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Precedence describes the order in which operations are performed in an arithmetic expression.

As in traditional mathematics, multiplication is done first:

```
let x = 100 + 50 * 3;
```

Try it Yourself »

When using parentheses, operations inside the parentheses are computed first:

```
let x = (100 + 50) * 3;
```

Try it Yourself »

When operators have the same precedence (like + and -), they are computed from left to right:

```
let x = 100 / 50 * 3;
```

Try it Yourself »

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Operator Precedence Values

Expressions in parentheses are computed **before** the rest of the expression
Functions are executed **before** the result is used in the rest of the expression

Val	Operator	Description	Example
18	( )	Expression Grouping	(100 + 50) * 3
17	.	Member Of	car.name
17	[]	Member Of	car["name"]
17	?.	Optional Chaining ES2020	x ?. y
17	()	Function Call	myFunction()
17	new	New with Arguments	new Date("June 5,2022")
16	new	New without Arguments	new Date()
		Increment Operators Postfix increments are executed **before** prefix increments	
15	++	Postfix Increment	i++
15	--	Postfix Decrement	i--
14	++	Prefix Increment	++i
14	--	Prefix Decrement	--i
		NOT Operators	
14	!	Logical NOT	!(x==y)
14	~	Bitwise NOT	~x
		Unary Operators	
14	+	Unary Plus	+x
14	-	Unary Minus	-x
14	typeof	Data Type	typeof x
14	void	Evaluate Void	void(0)
14	delete	Property Delete	delete myCar.color
		Arithmetic Operators Exponentiations are executed **before** multiplications Multiplications and divisions are executed **before** additions and subtractions	

13	**	Exponentiation ES2016	10 ** 2
12	*	Multiplication	10 * 5
12	/	Division	10 / 5
12	%	Division Remainder	10 % 5
11	+	Addition	10 + 5
11	-	Subtraction	10 - 5
11	+	Concatenation	"John" + "Doe"
		Shift Operators	
10	<<	Shift Left	x << 2
10	>>	Shift Right (signed)	x >> 2
10	>>>	Shift Right (unsigned)	x >>> 2
		Relational Operators	
9	in	Property in Object	"PI" in Math
9	instanceof	Instance of Object	x instanceof Array
		Comparison Operators	
9	<	Less than	x < y
9	<=	Less than or equal	x <= y
9	>	Greater than	x > y
9	>=	Greater than or equal	x >= Array
8	==	Equal	x == y
8	===	Strict equal	x === y
8	!=	Unequal	x != y
8	!==	Strict unequal	x !== y
		Bitwise Operators	
7	&	Bitwise AND	x & y
6	^	Bitwise XOR	x ^ y
5	|	Bitwise OR	x | y
		Logical Operators	

4	&&	Logical AND	x && y
3	||	Logical OR	x || y
3	??	Nullish Coalescing ES2020	x ?? y
		Conditional (ternary) Operator	
2	? :	Condition	? "yes" : "no"
		Assignment Operators Assignments are executed **after** other operations	
2	=	Simple Assignment	x = y
2	+=	Addition Assignment	x += y
2	-=	Subtraction Assignment	x -= y
2	*=	Multiplication Assignment	x *= y
2	**=	Exponentiation Assignment	x **= y
2	/=	Division Assignment	x /= y
2	%=	Remainder Assignment	x %= y
2	<<=	Left Shift Assignment	x <<= y
2	>>=	Right Shift Assignment	x >>= y
2	>>>=	Unsigned Right Shift	x >>>= y
2	&=	Bitwise AND Assignment	x &= y
2	|=	Bitwise OR Assignment	x |= y
2	^=	Bitwise XOR Assignment	x ^= y
2	&&=	Logical AND Assignment	x &= y
2	||=	Logical OR Assignment	x ||= y
2	:	Colon Assignment	x : 5
2	=>	Arrow	x => y
2	yield	Pause / Resume	yield x
2	yield*	Delegate	yield* x
2	...	Spread	...x
1	,	Comma	x , y

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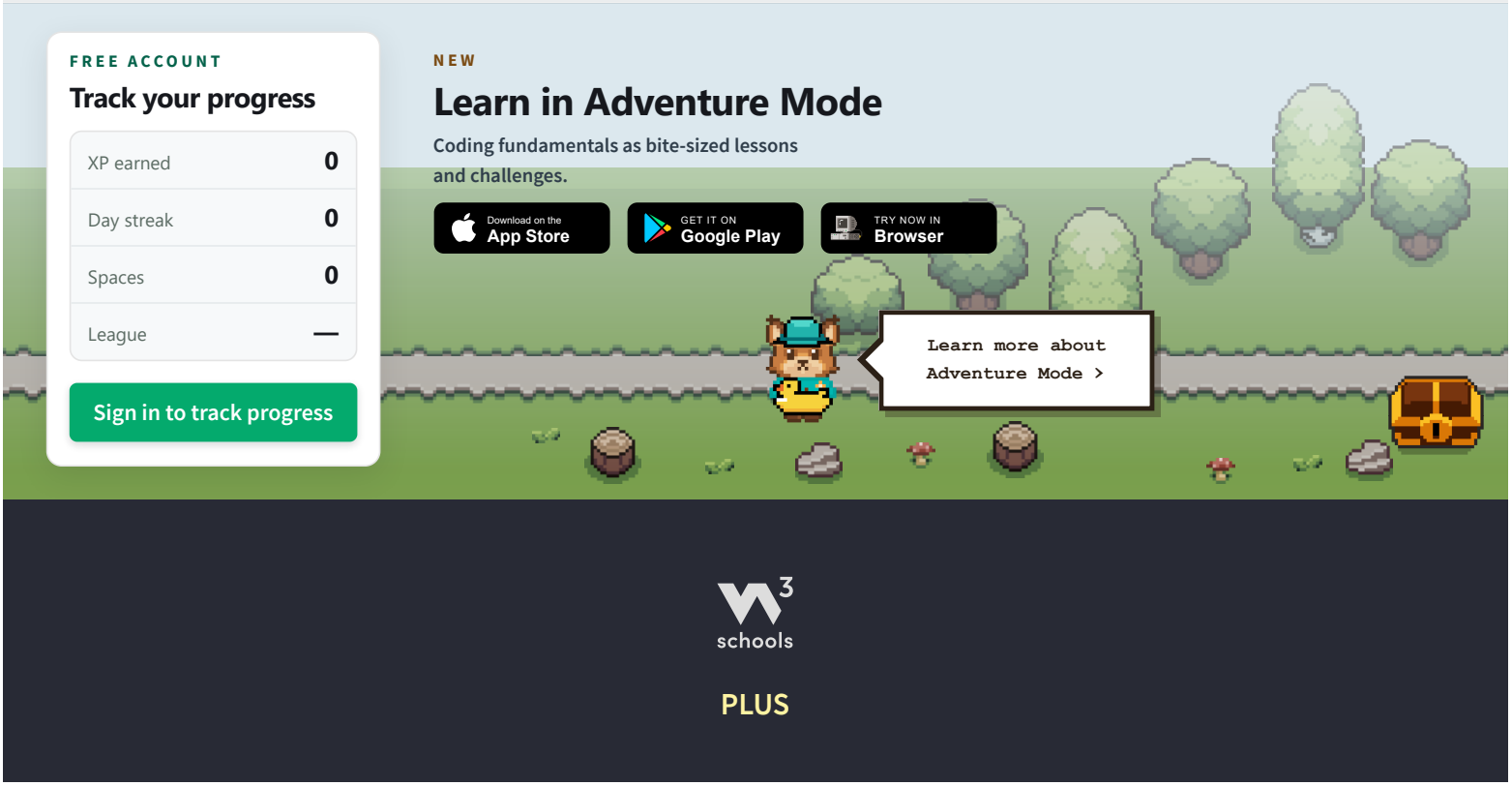
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JavaScript Array []

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Example

```
// Create an Array
const cars = ["Saab", "Volvo", "BMW"];
```

Try it Yourself »

More Examples Below !

Description

The `[ ]` constructor creates a new array object.

Syntax

```
[iterable]
```

Parameters

Parameter	Description

iterable	Required. An iterable object with values.

Return Value

Type	Description
Array	A new Array object.

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Array Iteration Methods

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More Examples

Example

Create an empty array and add values:

```
// Create an Array
const cars = [];

// Add Values to the Set
cars.push("Saab");
cars.push("Volvo");
cars.push("BMW");
```

Try it Yourself »

Example

Create an array with the new Array() method:

```
// Create an Array
const cars = new Array(["Saab", "Volvo", "BMW"]);
```

Try it Yourself »

Browser Support

`the [] array constructor` is an ECMAScript1 (JavaScript 1997) feature.

It is supported in all browsers:

Chrome	Edge	Firefox	Safari	Opera

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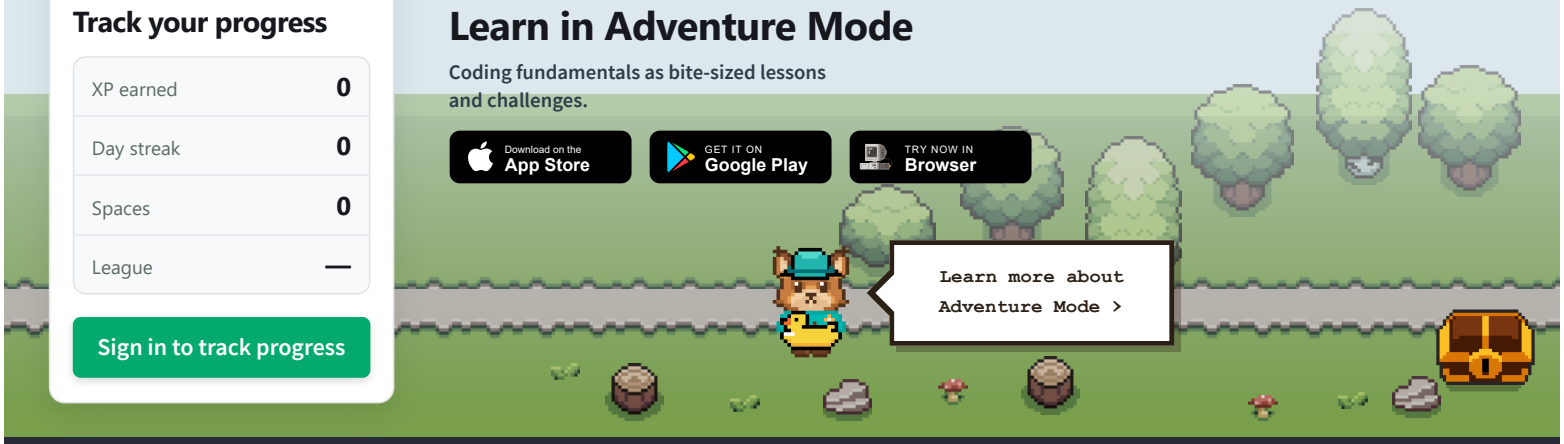
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JavaScript Array []

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Example

```
// Create an Array
const cars = ["Saab", "Volvo", "BMW"];
```

Try it Yourself »

More Examples Below !

Description

The `[ ]` constructor creates a new array object.

Syntax

```
[iterable]
```

Parameters

Parameter	Description

iterable	Required. An iterable object with values.

Return Value

Type	Description
Array	A new Array object.

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More Examples

Example

Create an empty array and add values:

```
// Create an Array
const cars = [];

// Add Values to the Set
cars.push("Saab");
cars.push("Volvo");
cars.push("BMW");
```

Try it Yourself »

Example

Create an array with the new Array() method:

```
// Create an Array
const cars = new Array(["Saab", "Volvo", "BMW"]);
```

Try it Yourself »

Browser Support

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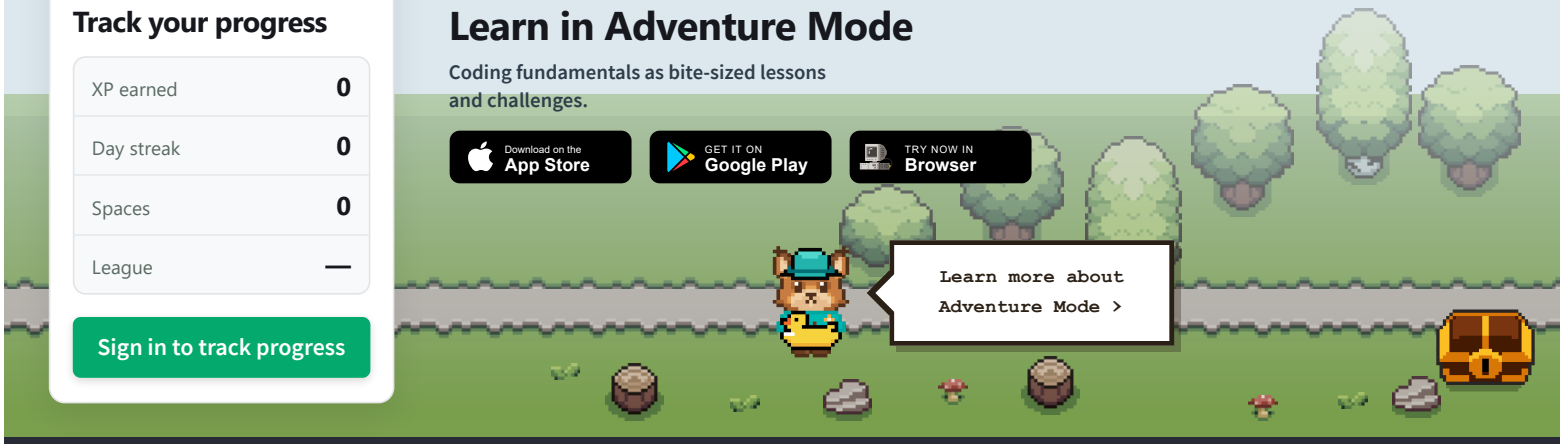
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JavaScript new Array()

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Examples

```
// Create a new Array
const cars = new Array("Saab", "Volvo", "BMW");
```

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```
const a = new Array();
const b = new Array(3);
const c = new Array("3");
const d = new Array("Saab", "Volvo", "BMW");
```

Try it Yourself »

More Examples Below !

Description

SHARE

The `new Array()` constructor creates a new Array object.

Syntax

```
new Array()
new Array(number)
new Array(non-number)
new Array(element1, ..., elementN)
new Array(array literal)
```

Parameters

Parameter	Description
none	Optional. Creates an empty array.
number	Optional. Creates an array with number of empty slots.
non-number	Optional. Creates an array with **non-number** as the only element.
element1, ..., elementN	Optional. **Comma separated values** to create a new new array from.
array literal	Optional. Creates an array from an **array literal**.

Parameters Explained

`new Array()` called with no arguments, creates an empty array.

Called with a number, it creates an array with that number of empty (null) elements.

Called with a non-number, it creates an array containing only that value.

Called with two or more values, it creates an array containing those values.

Called with an array literal, it creates an array from that literal.

Return Value

Type	Description
Array	A new Array object.

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More Examples

Example

Create an empty array and add values:

```
// Create a new Array
const cars = new Array();

// Add Values to the Set
cars.push("Saab");
cars.push("Volvo");
cars.push("BMW");
```

Try it Yourself »

Example

Create an array from a literal:

```
// Create a new Array
const cars = new Array(["Saab", "Volvo", "BMW"]);
```

Try it Yourself »

Example

Create an array with 10 empty (null) elements:

```
// Create an Array
const cars = new Array(10);
```

Try it Yourself »

Example

Create an array with one element:

```
// Create an Array
const cars = new Array("10");
```

Try it Yourself »

Example

Create an array without the new Array() method:

```
// Create an Array
const cars = ["Saab", "Volvo", "BMW"];
```

Try it Yourself »

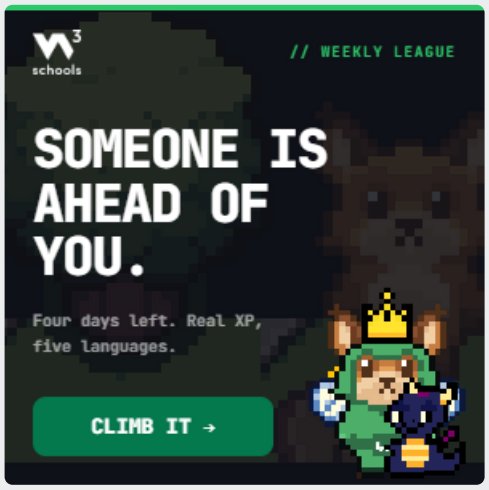
Browser Support

`new Array()` is an ECMAScript1 (JavaScript 1997) feature.

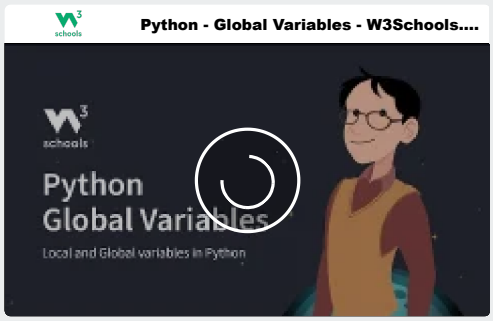
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JavaScript Array at()

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Examples

Get the third element of fruits:

```
const fruits = ["Banana", "Orange", "Apple", "Mango"];
let fruit = fruits.at(2);
```

Try it Yourself »

Get the third element of fruits:

```
const fruits = ["Banana", "Orange", "Apple", "Mango"];
let fruit = fruits[2];
```

Try it Yourself »

More examples below.

Description

SHARE

The `at()` method returns an indexed element from an array.

The `at()` method returns the same as `[]`.

The `at()` method is supported in all modern browsers since March 2022:

Note

Many languages allows `negative bracket indexing` like [-1] to access elements from the end of an object / array / string.

This is not possible in JavaScript, because [] is used for accessing both arrays and objects. obj[-1] refers to the value of key -1, not to the last property of the object.

The `at()` method was introduced in ES2022 to solve this problem.

Syntax

```
array.at(index)
```

Parameters

Parameter	Description
index	Optional. The index (position) of the array element to be returned. Default is 0. -1 returns the last element.

Return Value

Type	Description
Element	The element of the given position (index) in the array.

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Browser Support

JavaScript Array `at()` is supported in all browsers since March 2022:

				
Chrome 92	Edge 92	Firefox 90	Safari 15.4	Opera 78
Apr 2021	Jul 2021	Jul 2021	Mar 2022	Aug 2021

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More Examples

Get the first element of fruits:

```
const fruits = ["Banana", "Orange", "Apple", "Mango"];
let fruit = fruits.at();
```

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Get the last element of fruits:

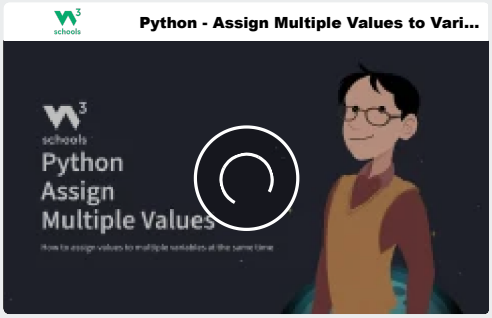
```
const fruits = ["Banana", "Orange", "Apple", "Mango"];
let fruit = fruits.at(-1);
```

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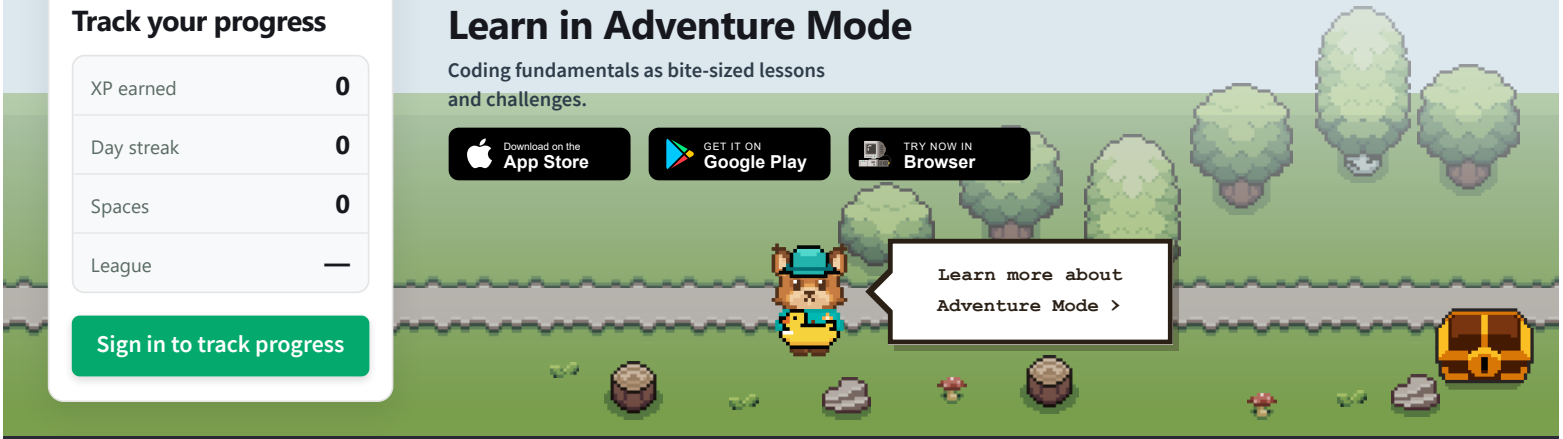
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JavaScript Array concat()

‹ Previous JavaScript Array Reference Next ›

Examples

Join two arrays:

```
const arr1 = ["Cecilie", "Lone"];
const arr2 = ["Emil", "Tobias", "Linus"];
const children = arr1.concat(arr2);
```

Try it Yourself »

Join three arrays:

```
const arr1 = ["Cecilie", "Lone"];
const arr2 = ["Emil", "Tobias", "Linus"];
const arr3 = ["Robin"];
const children = arr1.concat(arr2, arr3);
```

Try it Yourself »

More examples below.

Description

SHARE

The `concat()` method concatenates (joins) two or more arrays.

The `concat()` method returns a new array, containing the joined arrays.

The `concat()` method does not change the existing arrays.

See Also:

The join() Method

The slice() Method

The splice() Method

The copyWithin() Method

Syntax

```
array1.concat(array2, array3, ..., arrayX)
```

Parameters

Parameter	Description
array1,...	Required. The array(s) to be concatenated.

Return Value

Type	Description
Array	The content from the joined arrays.

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More Examples

Concatenate strings and numbers:

```
const arr1 = ["Cecilie", "Lone"];
const arr2 = [1, 2, 3];
const arr3 = arr1.concat(arr2);
```

Try it Yourself »

Concatenate nested arrays:

```
const arr1 = [1, 2, [3, 4]];
const arr2 = [[5, 6], 7, 8];
const arr3 = arr1.concat(arr2);
```

Try it Yourself »

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Array Search Methods

Array Sort Methods

Array Iteration Methods

Browser Support

`concat()` is an ECMAScript1 (JavaScript 1997) feature.

It is supported in all browsers:

				
Chrome	Edge	Firefox	Safari	Opera

‹ Previous JavaScript Array Reference Next ›

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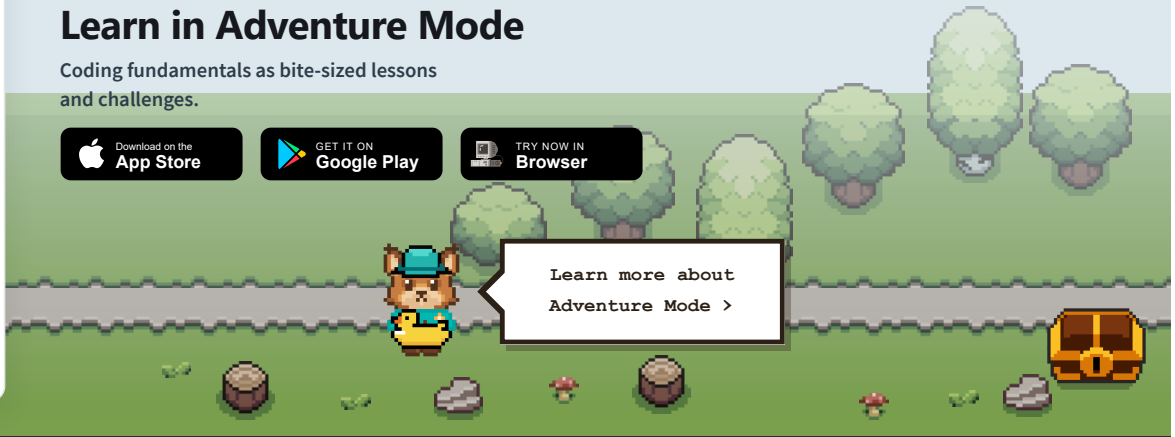
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JavaScript Array constructor

‹ Previous JavaScript Array Reference Next ›

Example

Get the array constructor:

```
const fruits = ["Banana", "Orange", "Apple", "Mango"];
let text = fruits.constructor;
```

Try it Yourself »

Description

SHARE

The `constructor` property returns the function that created the Array prototype.

For JavaScript arrays the `constructor` property returns:

function Array() { [native code] }

Syntax

```
array.constructor
```

Return Value

```
function Array() { [native code] }
```

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Array Search Methods

Array Sort Methods

Array Iteration Methods

Browser Support

`constructor` is an ECMAScript1 (JavaScript 1997) feature.

It is supported in all browsers:

Chrome	Edge	Firefox	Safari	Opera

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JavaScript Array Reference

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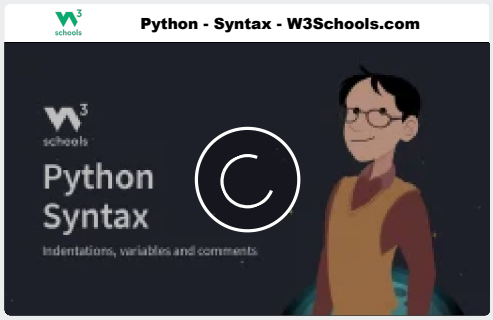
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JavaScript Array copyWithin()

‹ Previous JavaScript Array Reference Next ›

Examples

Copy to index 2, all elements from index 0:

```
const fruits = ["Banana", "Orange", "Apple", "Mango"];
fruits.copyWithin(2, 0);
```

Try it Yourself »

Copy to index 2, the elements from index 0 to 2:

```
const fruits = ["Banana", "Orange", "Apple", "Mango", "Kiwi"];
fruits.copyWithin(2, 0, 2);
```

Try it Yourself »

Description

SHARE

The `copyWithin()` method copies array elements to another position in an array.

The `copyWithin()` method overwrites the existing values.

The `copyWithin()` method does not add items to the array.

Syntax

```
array.copyWithin(target, start, end)
```

Parameters

Parameter	Description
target	Required. The index (position) to copy the elements to.
start	Optional. The start index (position). Default is 0.
end	Optional. The end index (position). Default is the array length.

Return Value

Type	Description
Array	The changed array.

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Browser Support

`copyWithin()` is an ECMAScript6 (ES6 2015) feature.

JavaScript 2015 is supported in all browsers since **June 2017**:

Chrome	Edge	Firefox	Safari	Opera
51	15	54	10	38
May 2016	Apr 2017	Jun 2017	Sep 2016	Jun 2016

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JavaScript Array entries()

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Example

```
// Create an Array
const fruits = ["Banana", "Orange", "Apple", "Mango"];

// Create an Iterator
const list = fruits.entries();

// List the Entries
let text = "";
for (let x of list) {
  text += x;
}
```

Try it Yourself »

More Examples Below!

Description

SHARE

The `entries()` method returns an Iterator object with the key/value pairs from an array:

[0, "Banana"]
[1, "Orange"]
[2, "Apple"]
[3, "Mango"]

The `entries()` method does not change the original array.

Array Iteration Methods:

The Array entries() Method

The Array every() Method

The Array filter() Method

The Array forEach() Method

The Array keys() Method

The Array map() Method

Syntax

```
array.entries()
```

Parameters

NONE

Return Value

Type	Description
Iterable	An Iterable object with the key/value pairs from the array.

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More Examples

Example

Iterate directly over the Iterator:

```
// Create an Array
const fruits = ["Banana", "Orange", "Apple", "Mango"];

// List the Entries
let text = "";
for (let x of fruits.entries()) {
  text += x;
}
```

Try it Yourself »

Example

Use the built in Object.entries() method:

```
// Create an Array
const fruits = ["Banana", "Orange", "Apple", "Mango"];

// List the Entries
let text = "";
for (let x of Object.entries(fruits)) {
  text += x;
}
```

Try it Yourself »

Note

It is not a good practice to save an iterator.

The iterator has a next() method to access each element one at a time.

As soon as you start using it, it cannot be reset or restarted.

Example

Use the next() method of the iterator:

```
// Create an Array
const fruits = ["Banana", "Orange", "Apple", "Mango"];

// Create an Interator
const list = fruits.entries();
```

```
let text = list.next().value + " " + list.next().value;
```

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Browser Support

`entries()` is an ECMAScript6 (ES6 2015) feature.

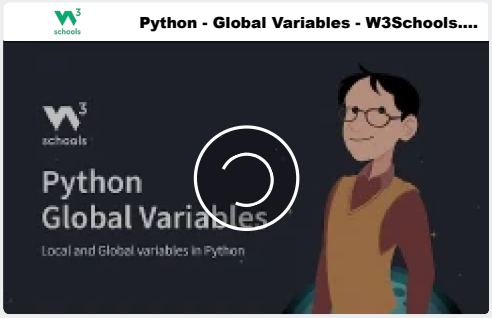
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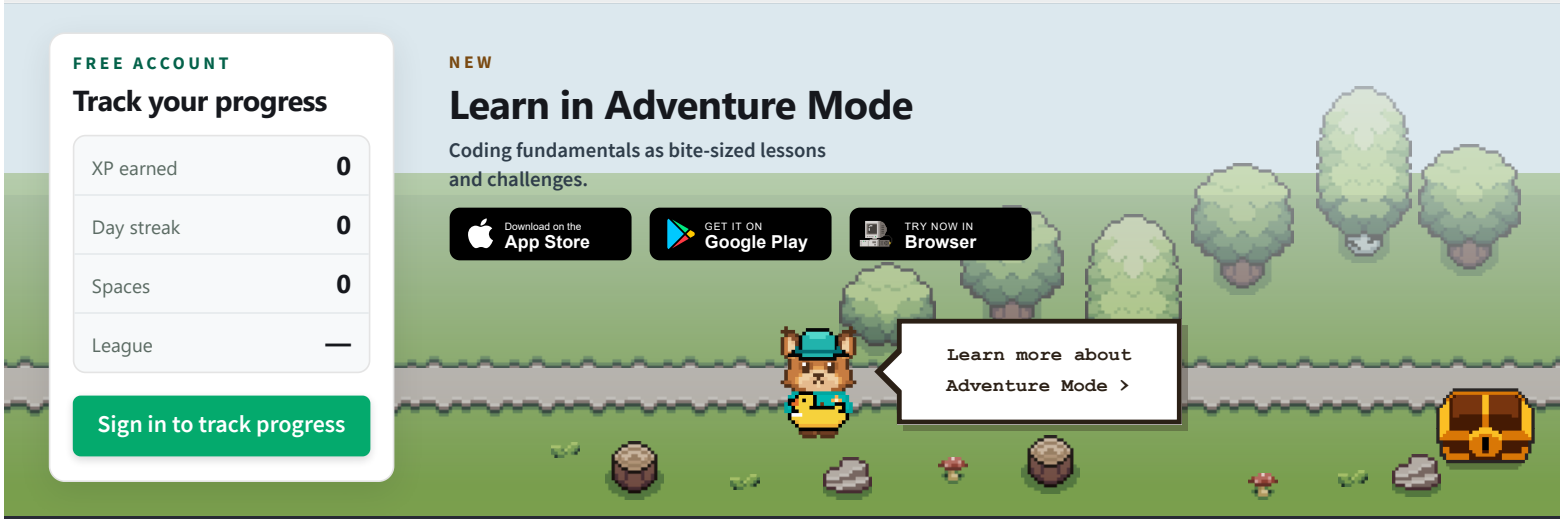
				
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May 2016	Apr 2017	Jun 2017	Sep 2016	Jun 2016

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JavaScript Array every()

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Example 1

```
// Create an Array
const ages = [32, 33, 16, 40];

// Create a Test Function
function checkAge(age) {
  return age > 18;
}

// Are all ages over 18?
ages.every(checkAge);
```

Try it Yourself »

More examples below.

Description

SHARE

The `every()` method executes a function for each array element.

The `every()` method returns `true` if the function returns true for all elements.

The `every()` method returns `false` if the function returns false for one element.

The `every()` method does not execute the function for empty elements.

The `every()` method does not change the original array

Array Iteration Methods:

The Array entries() Method

The Array every() Method

The Array filter() Method

The Array forEach() Method

The Array keys() Method

The Array map() Method

Syntax

```
array.every(function(currentValue, index, arr), thisValue)
```

Parameters

Parameter	Description
function()	Required. A function to be run for each element in the array.
currentValue	Required. The value of the current element.
index	Optional. The index of the current element.
arr	Optional. The array of the current element.
thisValue	Optional. Default `undefined`. A value passed to the function as its `this` value.

Return Value

Type	Description
Boolean	`true` if all elements pass the test, otherwise `false`.

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More Examples

Check if all answers are the same:

```
const survey = [
  { name: "Steve", answer: "Yes"},
  { name: "Jessica", answer: "Yes"},
  { name: "Peter", answer: "Yes"},
  { name: "Elaine", answer: "No"}
];

let result = survey.every(isSameAnswer);

function isSameAnswer(el, index, arr) {
  if (index === 0) {
    return true;
  } else {
    return (el.answer === arr[index - 1].answer);
  }
}
```

Try it Yourself »

Check if all values are over a specific number:

```
<p><input type="number" id="ageToCheck" value="18"></p>

<button onclick="myFunction()">Try it</button>

<p id="demo"></p>

<script>
const ages = [32, 33, 12, 40];

function checkAge(age) {
  return age > document.getElementById("ageToCheck").value;
}

function myFunction() {
  document.getElementById("demo").innerHTML = ages.every(checkAge);
}
</script>
```

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Array Iteration Methods

Browser Support

`every()` is an ECMAScript5 (ES5 2009) feature.

JavaScript 2009 is supported in all browsers since **July 2013**:

Chrome	IE/Edge	Firefox	Safari	Opera
23	11	21	6	15
Sep 2012	Sep 2012	Apr 2013	Jul 2012	Jul 2013

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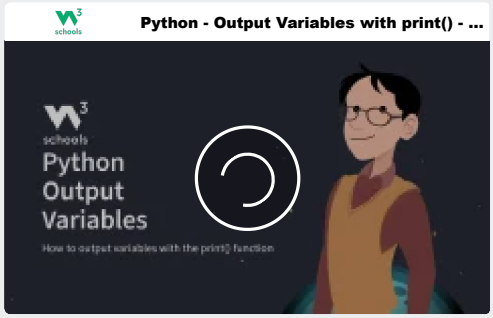
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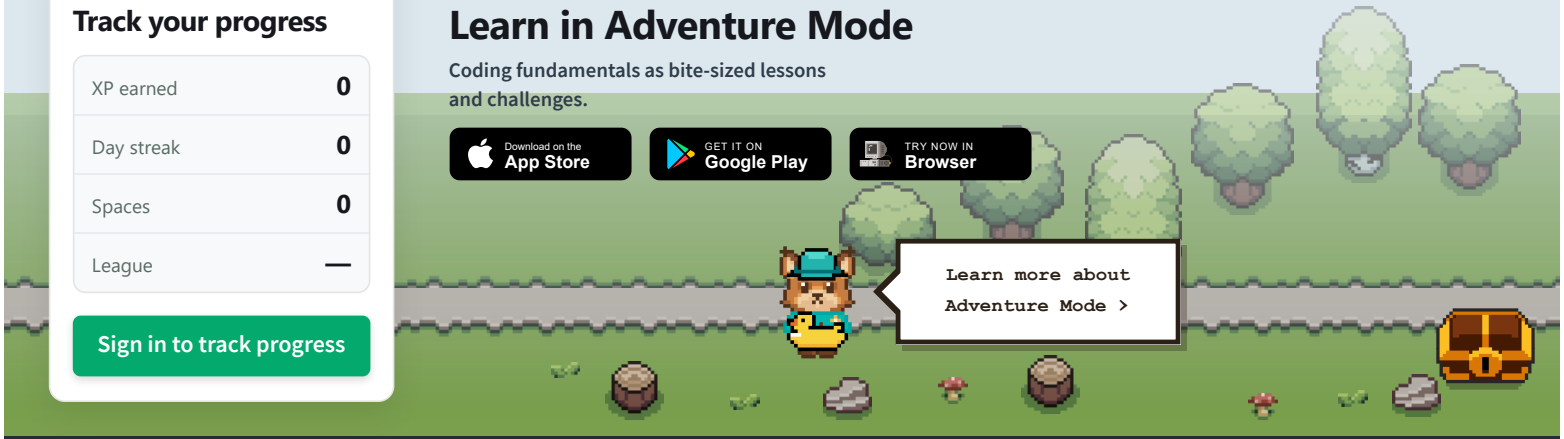
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JavaScript Array fill()

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Examples

Fill all the elements with a value:

```
const fruits = ["Banana", "Orange", "Apple", "Mango"];
fruits.fill("Kiwi");
```

Try it Yourself »

Fill the last two elements:

```
const fruits = ["Banana", "Orange", "Apple", "Mango"];
fruits.fill("Kiwi", 2, 4);
```

Try it Yourself »

Description

SHARE

The `fill()` method fills specified elements in an array with a value.

The `fill()` method overwrites the original array.

Start and end position can be specified. If not, all elements will be filled.

Syntax

```
array.fill(value, start, end)
```

Parameters

Parameter	Description
value	Required. The value to fill in.
start	Optional. The start index (position). Default is 0.
end	Optional. The stop index (position). Default is array length.

Return Value

Type	Description
Array	The filled array.

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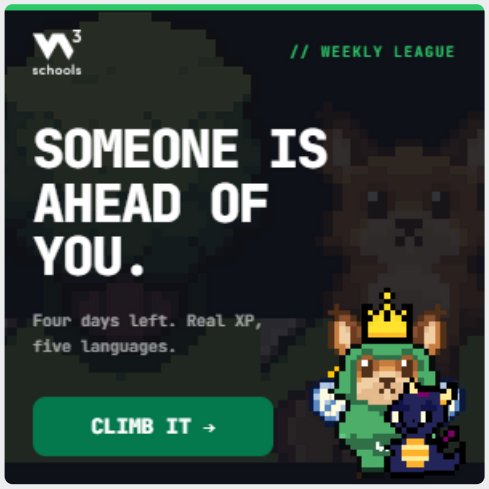
Browser Support

`fill()` is an ECMAScript6 (ES6 2015) feature.

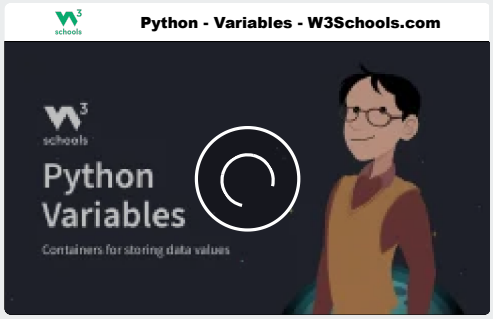
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JavaScript Array filter()

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Example 1

Return an array of all values in ages[] that are 18 or over:

```
const ages = [32, 33, 16, 40];
const result = ages.filter(checkAdult);

function checkAdult(age) {
  return age >= 18;
}
```

Try it Yourself »

Description

SHARE

The `filter()` method creates a new array filled with elements that pass a test provided by a function.

The `filter()` method does not execute the function for empty elements.

The `filter()` method does not change the original array.

Array Iteration Methods:

The Array entries() Method

The Array every() Method

The Array filter() Method

The Array forEach() Method

The Array keys() Method

The Array map() Method

Syntax

```
array.filter(function(currentValue, index, arr), thisValue)
```

Parameters

Parameter	Description
function()	Required. A function to run for each array element.
currentValue	Required. The value of the current element.
index	Optional. The index of the current element.
arr	Optional. The array of the current element.
thisValue	Optional. Default `undefined` A value passed to the function as its `this` value.

Return Value

Type	Description
Array	An array of elements that pass the test. An empty array if no elements pass the test.

Example 2

Return the values in ages[] that are over a specific number:

```
<p><input type="number" id="ageToCheck" value="30"></p>

<button onclick="myFunction()">Try it</button>

<p id="demo"></p>

<script>
const ages = [32, 33, 12, 40];

function checkAge(age) {
  return age > document.getElementById("ageToCheck").value;
}

function myFunction() {
  document.getElementById("demo").innerHTML = ages.filter(checkAge);
}
</script>
```

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Browser Support

`filter()` is an ECMAScript5 (ES5 2009) feature.

JavaScript 2009 is supported in all browsers since **July 2013**:

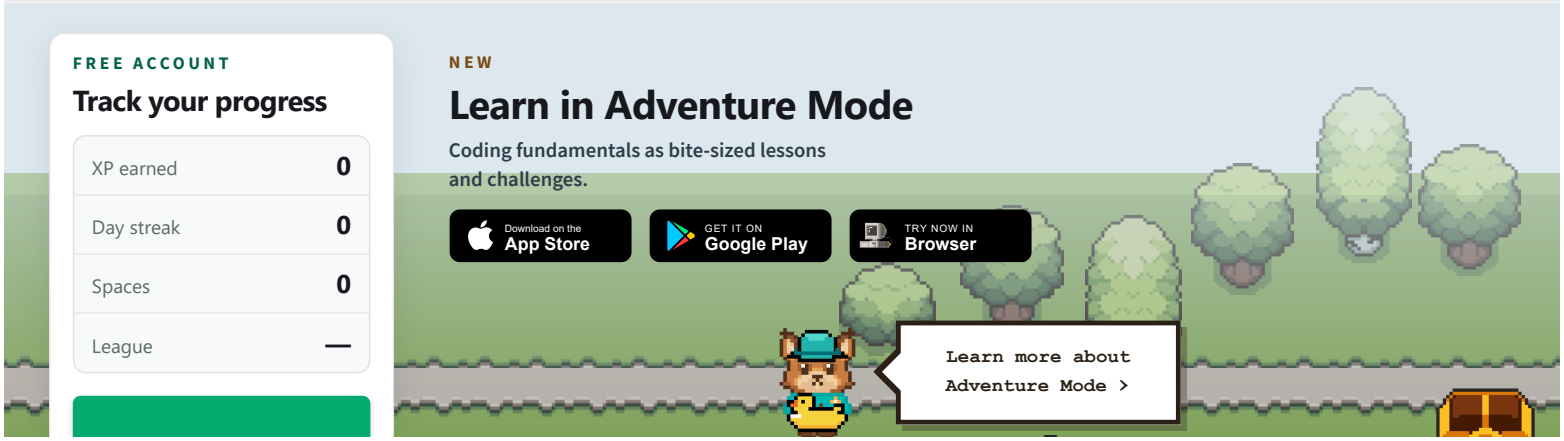
Chrome 23	IE/Edge 11	Firefox 21	Safari 6	Opera 15
Sep 2012	Sep 2012	Apr 2013	Jul 2012	Jul 2013

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JavaScript Array find()

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Example 1

Find the value of the first element with a value over 18:

```
const ages = [3, 10, 18, 20];

function checkAge(age) {
  return age > 18;
}

function myFunction() {
  document.getElementById("demo").innerHTML = ages.find(checkAge);
}
```

Try it Yourself »

Description

SHARE

The `find()` method returns the value of the first element that passes a test.

The `find()` method executes a function for each array element.

The `find()` method returns `undefined` if no elements are found.

The `find()` method does not execute the function for empty elements.

The `find()` method does not change the original array.

Array Find Methods:

Method	Finds
includes()	Returns true if an array contains a specified value
indexOf()	The index of the first element with a specified value
lastIndexOf()	The index of the last element with a specified value
find()	The value of the first element that passes a test
findIndex()	The index of the first element that passes a test
findLast()	The value of the last element that passes a test
findLastIndex()	The index of the last element that passes a test

Syntax

```
array.find(function(currentValue, index, arr),thisValue)
```

Parameters

function()	Required. A function to run for each array element.
currentValue	Required. The value of the current element.
index	Optional. The index of the current element.
arr	Optional. The array of the current element.
thisValue	Optional. Default `undefined`. A value passed to the function as its `this` value.

Return Value

Type	Description
A value	The value of the first element that pass the test. Otherwise it returns `undefined`.

Example 2

Find the value of the first element with a value above a specific number:

```
<p><input type="number" id="ageToCheck" value="18"></p>

<button onclick="myFunction()">Try it</button>

<p id="demo"></p>

<script>
const ages = [4, 12, 16, 20];

function checkAge(age) {
  return age > document.getElementById("ageToCheck").value;
}

function myFunction() {
  document.getElementById("demo").innerHTML = ages.find(checkAge);
}
</script>
```

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Browser Support

`find()` is an ECMAScript6 (ES6 2015) feature.

JavaScript 2015 is supported in all browsers since **June 2017**:

Chrome	Edge	Firefox	Safari	Opera
51	15	54	10	38
May 2016	Apr 2017	Jun 2017	Sep 2016	Jun 2016

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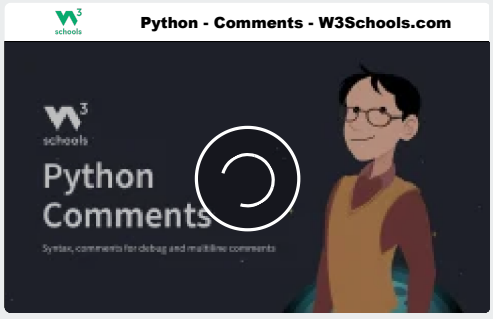
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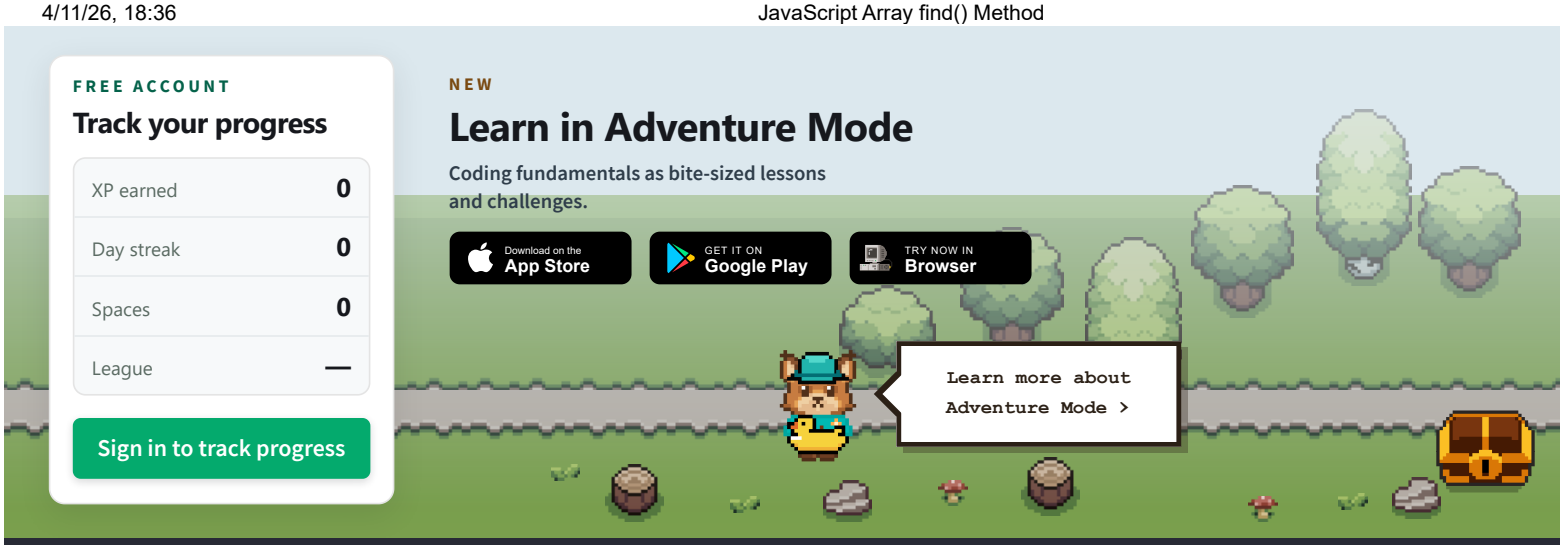
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JavaScript Array findIndex()

‹ Previous JavaScript Array Reference Next ›

Example 1

Find the first element with a value over 18:

```
const ages = [3, 10, 18, 20];

ages.findIndex(checkAge);

function checkAge(age) {
  return age > 18;
}
```

Try it Yourself »

Description

SHARE

The `findIndex()` method executes a function for each array element.

The `findIndex()` method returns the index (position) of the first element that passes a test.

The `findIndex()` method returns -1 if no match is found.

The `findIndex()` method does not execute the function for empty array elements.

The `findIndex()` method does not change the original array.

Array Find Methods:

Method	Finds
includes()	Returns true if an array contains a specified value
indexOf()	The index of the first element with a specified value
lastIndexOf()	The index of the last element with a specified value
find()	The value of the first element that passes a test
findIndex()	The index of the first element that passes a test
findLast()	The value of the last element that passes a test
findLastIndex()	The index of the last element that passes a test

Syntax

```
array.findIndex(function(currentValue, index, arr), thisValue)
```

Parameters

Parameter	Description
function()	Required. A function to be run for each array element.
currentValue	Required. The value of the current element.
index	Optional. The index of the current element.
arr	Optional. The array of the current element.
thisValue	Optional. Default `undefined`. A value passed to the function as its `this` value.

Return Value

Type	Description

Number	The index of the first element that passes the test. Otherwise -1.

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More Examples

Find the first element with a value above an input value:

```
<p><input type="number" id="toCheck" value="18"></p>

<button onclick="myFunction()">Test</button>

<p>Any values above: <span id="demo"></span></p>

<script>
const numbers = [4, 12, 16, 20];

function checkValue(x) {
  return x > document.getElementById("toCheck").value;
}

function myFunction() {
  document.getElementById("demo").innerHTML = numbers.findIndex(checkValue);
}
</script>
```

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Browser Support

`findIndex()` is an ECMAScript6 (ES6 2015) feature.

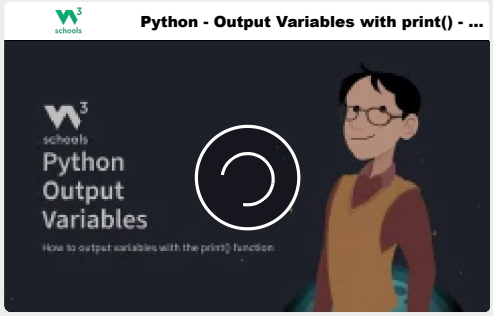
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51	15	54	10	38
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JavaScript Array findLast()

‹ Previous JavaScript Array Reference Next ›

Example 1

Find the value of the last element with a value over 18:

```
const ages = [3, 10, 18, 20];

function checkAge(age) {
  return age > 18;
}

function myFunction() {
  document.getElementById("demo").innerHTML = ages.findLast(checkAge);
}
```

Try it Yourself »

Description

SHARE

The `findLast()` method returns the value of the last element that passes a test.

The `findLast()` method executes a function for each array element.

The `findLast()` method returns `undefined` if no elements are found.

The `findLast()` method does not execute the function for empty elements.

The `findLast()` method does not change the original array.

Array Find Methods:

Method	Finds
includes()	Returns true if an array contains a specified value
indexOf()	The index of the first element with a specified value
lastIndexOf()	The index of the last element with a specified value
find()	The value of the first element that passes a test
findIndex()	The index of the first element that passes a test
findLast()	The value of the last element that passes a test
findLastIndex()	The index of the last element that passes a test

Syntax

```
array.findLast(function(currentValue, index, arr),thisValue)
```

Parameters

function()	Required. A function to run for each array element.
currentValue	Required. The value of the current element.
index	Optional. The index of the current element.
arr	Optional. The array of the current element.
thisValue	Optional. Default `undefined`. A value passed to the function as its `this` value.

Return Value

Type	Description
A value	The value of the last element that pass the test. Otherwise it returns `undefined`.

Example 2

Find the value of the last element with a value above a specific number:

```
<p><input type="number" id="ageToCheck" value="18"></p>

<button onclick="myFunction()">Try it</button>

<p id="demo"></p>

<script>
const ages = [4, 12, 16, 20];

function checkAge(age) {
  return age > document.getElementById("ageToCheck").value;
}

function myFunction() {
  document.getElementById("demo").innerHTML = ages.findLast(checkAge);
}
</script>
```

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Browser Support

`findLast()` is a JavaScript 2023 feature.

ES 2023 is supported in all modern browsers since **July 2023**:

Chrome	Edge	Firefox	Safari	Opera
110	110	115	16.4	96
Feb 2023	Feb 2023	Jul 2023	Mar 2023	May 2023

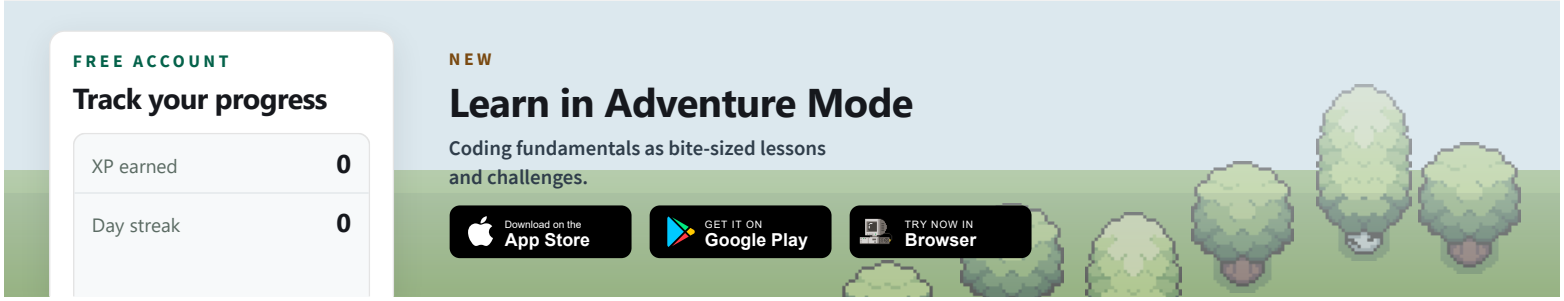
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JavaScript Array findLastIndex()

‹ Previous JavaScript Array Reference Next ›

Example 1

Find the last element with a value over 18:

```
const ages = [3, 10, 18, 20];

ages.findLastIndex(checkAge);

function checkAge(age) {
  return age > 18;
}
```

Try it Yourself »

Description

SHARE

The `findLastIndex()` method executes a function for each array element.

The `findLastIndex()` method returns the index (position) of the last element that passes a test.

The `findLastIndex()` method returns -1 if no match is found.

The `findLastIndex()` method does not execute the function for empty array elements.

The `findLastIndex()` method does not change the original array.

Array Find Methods:

Method	Finds
includes()	Returns true if an array contains a specified value
indexOf()	The index of the first element with a specified value
lastIndexOf()	The index of the last element with a specified value
find()	The value of the first element that passes a test
findIndex()	The index of the first element that passes a test
findLast()	The value of the last element that passes a test
findLastIndex()	The index of the last element that passes a test

Syntax

```
array.findLastIndex(function(currentValue, index, arr), thisValue)
```

Parameters

Parameter	Description
function()	Required. A function to be run for each array element.
currentValue	Required. The value of the current element.
index	Optional. The index of the current element.
arr	Optional. The array of the current element.
thisValue	Optional. Default `undefined`. A value passed to the function as its `this` value.

Return Value

Type	Description

Number	The index of the last element that passes the test. Otherwise -1.

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More Examples

Find the last element with a value above an input value:

```
<p><input type="number" id="toCheck" value="18"></p>

<button onclick="myFunction()">Test</button>

<p>Any values above: <span id="demo"></span></p>

<script>
const numbers = [4, 12, 16, 20];

function checkValue(x) {
  return x > document.getElementById("toCheck").value;
}

function myFunction() {
  document.getElementById("demo").innerHTML = numbers.findLastIndex(checkValue);
}
</script>
```

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Browser Support

`findLastIndex()` is a JavaScript 2023 feature.

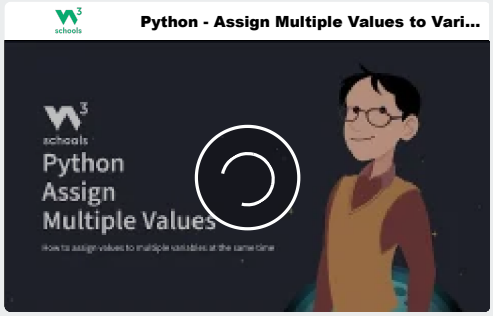
ES 2023 is supported in all modern browsers since **July 2023**:

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110	110	115	16.4	96
Feb 2023	Feb 2023	Jul 2023	Mar 2023	May 2023

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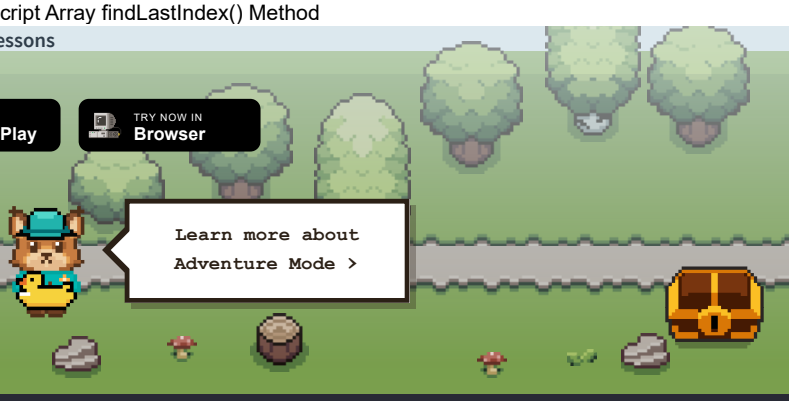


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JavaScript Array flat()

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Examples

Create a new array with one sub-array element concatenated:

```
const myArr = [1, 2, [3, [4, 5, 6], 7], 8];

const newArr = myArr.flat(1);

let text = JSON.stringify(newArr);
```

Try it Yourself »

Create a new array with two sub-array elements concatenated:

```
const myArr = [1, 2, [3, [4, 5, 6], 7], 8];

const newArr = myArr.flat(2);

let text = JSON.stringify(newArr);
```

Try it Yourself »

Note

SHARE

Arrays displayed in HTML are automatically converted to strings.

To display the full structure (including brackets), use JSON.stringify().

Description

The `flat()` method concatenates sub-array elements.

Syntax

```
array.flat(depth)
```

Parameters

Parameter	Description
depth	Optional. How deep a nested array should be flattened. Default is 1.

Return Value

Type	Description
An array	The flattened array.

See Also:

The Array map() Method

The Array flatMap() Method

The Array filter() Method

The Array forEach() Method

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More Examples

Example

```
const myArr = [[1,2],[3,4],[5,6]];
const newArr = myArr.flat();
```

Try it Yourself »

Example

```
const myArr = [1, 2, [3, [4, 5, 6], 7], 8];
const newArr = myArr.flat();
```

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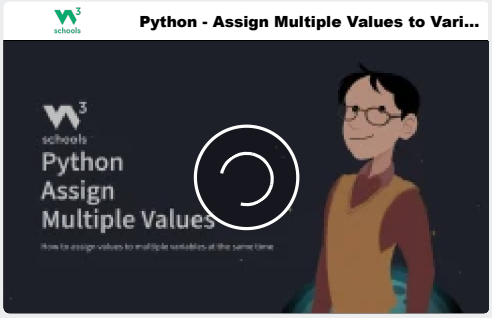
JavaScript Array `flat()` is supported in all modern browsers since January 2020:

				
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JavaScript Array flatMap()

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Example

```
const myArr = [1, 2, 3, 4, 5, 6];
const newArr = myArr.flatMap(x => [x, x * 10]);
```

Try it Yourself »

Description

SHARE

The `flatMap()` method maps all array elements and creates a new flat array.

`flatMap()` creates a new array from calling a function for every array element.

`flatMap()` does not execute the function for empty elements.

`flatMap()` does not change the original array.

See Also:

The Array map() Method

The Array flat() Method

The Array filter() Method

The Array forEach() Method

Syntax

```
array.flatMap(function(currentValue, index, arr), thisValue)
```

Parameters

Parameter	Description
function()	Required. A function to be run for each array element.
currentValue	Required. The value of the current element.
index	Optional. The index of the current element.
arr	Optional. The array of the current element.
thisValue	Optional. Default value `undefined`. A value passed to the function to be used as its `this` value.

Return Value

Type	Description
An array	An array with the elements as a result of a callback function and then flattened.

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Browser Support

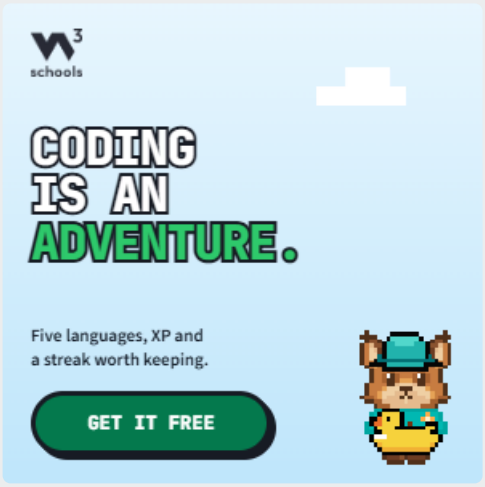
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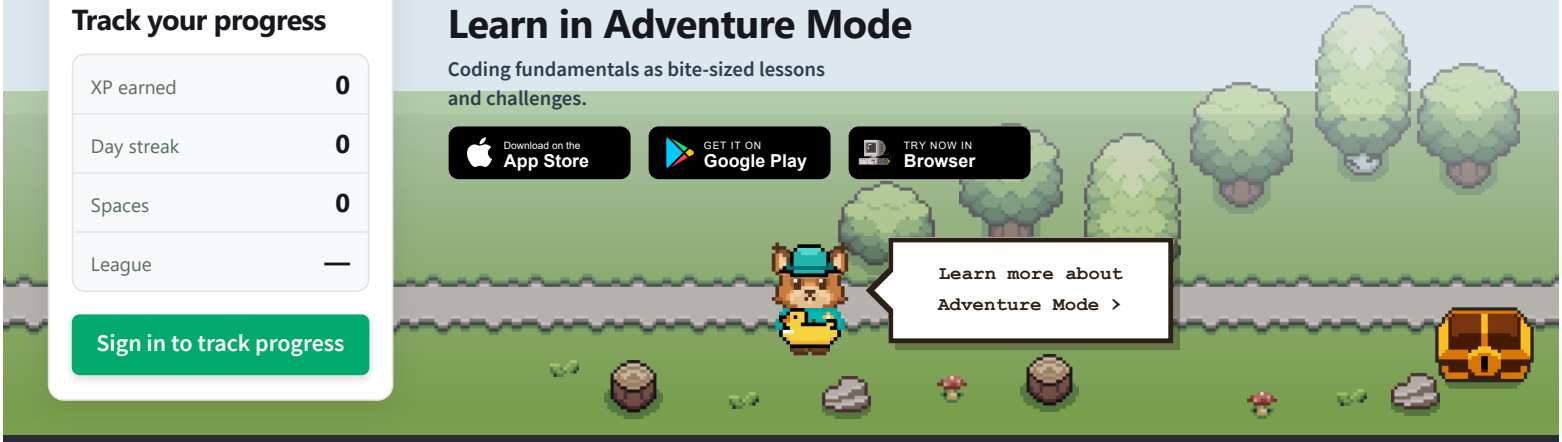


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JavaScript Array forEach()

‹ Previous JavaScript Array Reference Next ›

Example 1

Calls a function for each element in fruits:

```
const fruits = ["apple", "orange", "cherry"];
fruits.forEach(myFunction);
```

Try it Yourself »

Description

SHARE

The `forEach()` method calls a function for each element in an array.

The `forEach()` method is not executed for empty elements.

Array Iteration Methods:

The Array entries() Method

The Array every() Method

The Array filter() Method

The Array forEach() Method

The Array keys() Method

The Array map() Method

Syntax

```
array.forEach(function(currentValue, index, arr), thisValue)
```

Parameters

function()	Required. A function to run for each array element.
currentValue	Required. The value of the current element.
index	Optional. The index of the current element.
arr	Optional. The array of the current element.
thisValue	Optional. Default `undefined`. A value passed to the function as its `this` value.

Return Value

undefined

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More Examples

Compute the sum:

```
let sum = 0;
const numbers = [65, 44, 12, 4];
numbers.forEach(myFunction);

function myFunction(item) {
```

```
  sum += item;
}
```

Try it Yourself »

Multiply each element:

```
const numbers = [65, 44, 12, 4];
numbers.forEach(myFunction)

function myFunction(item, index, arr) {
  arr[index] = item * 10;
}
```

Try it Yourself »

Array Tutorials:

Array Tutorial

Array Const

Basic Array Methods

Array Search Methods

Array Sort Methods

Array Iteration Methods

Browser Support

`forEach()` is an ECMAScript5 (ES5 2009) feature.

JavaScript 2009 is supported in all browsers since **July 2013**:

				
Chrome 23	IE/Edge 11	Firefox 21	Safari 6	Opera 15
Sep 2012	Sep 2012	Apr 2013	Jul 2012	Jul 2013

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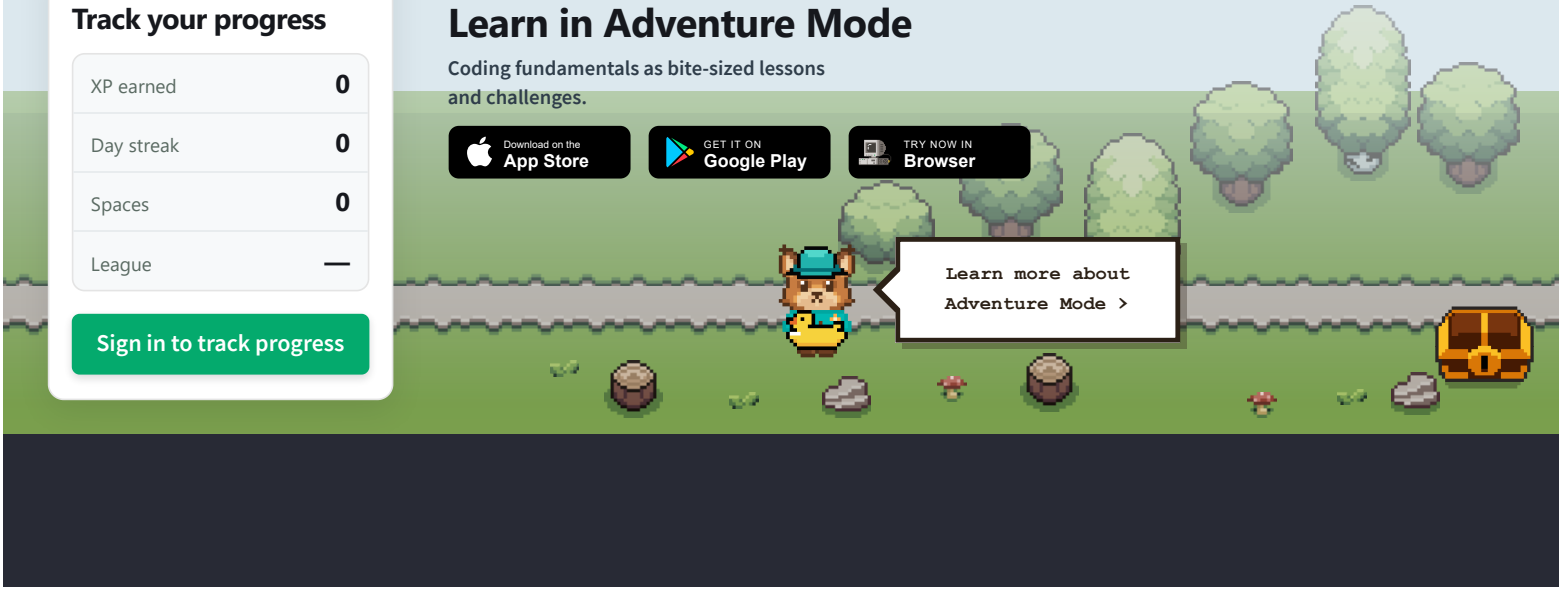
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JavaScript Array.from()

‹ Previous JavaScript Array Reference Next ›

Example

Create an array from a string:

```
let text = "ABCDEFG";
Array.from(text);
```

Try it Yourself »

More examples below !

Description

SHARE

The `Array.from()` method returns an array from any object with a length property.

The `Array.from()` method returns an array from any iterable object.

Note

Array.from() is a static property of the JavaScript Array object.

You can only use it as Array.from().

Using x.from(), where x is an array will return undefined.

Syntax

```
Array.from(object, mapFunction, thisValue)
```

Parameters

Parameter	Description
object	Required. The object to convert to an array.
mapFunction	Optional. A map function to call on each item.
thisValue	Optional. A value to use as `this` for the mapFunction

Return Value

Type	Description
An array	The values from the iterable object.

More Examples

Example

Create an array from an array:

```
const myNumbers = [1,2,3,4];
const myArr = Array.from(myNumbers, (x) => x * 2);
```

Try it Yourself »

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Array Tutorials:

Array Tutorial

Array Const

Basic Array Methods

Array Search Methods

Array Sort Methods

Array Iteration Methods

Browser Support

`Array.from()` is an ECMAScript6 (ES6 2015) feature.

JavaScript 2015 is supported in all browsers since **June 2017**:

Chrome	Edge	Firefox	Safari	Opera
51	15	54	10	38
May 2016	Apr 2017	Jun 2017	Sep 2016	Jun 2016

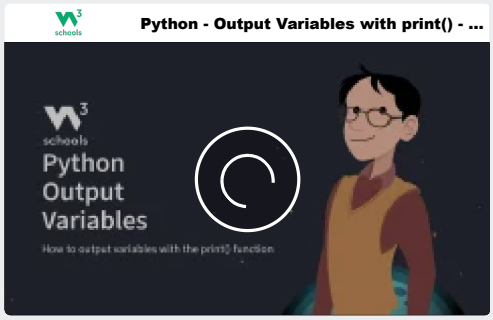
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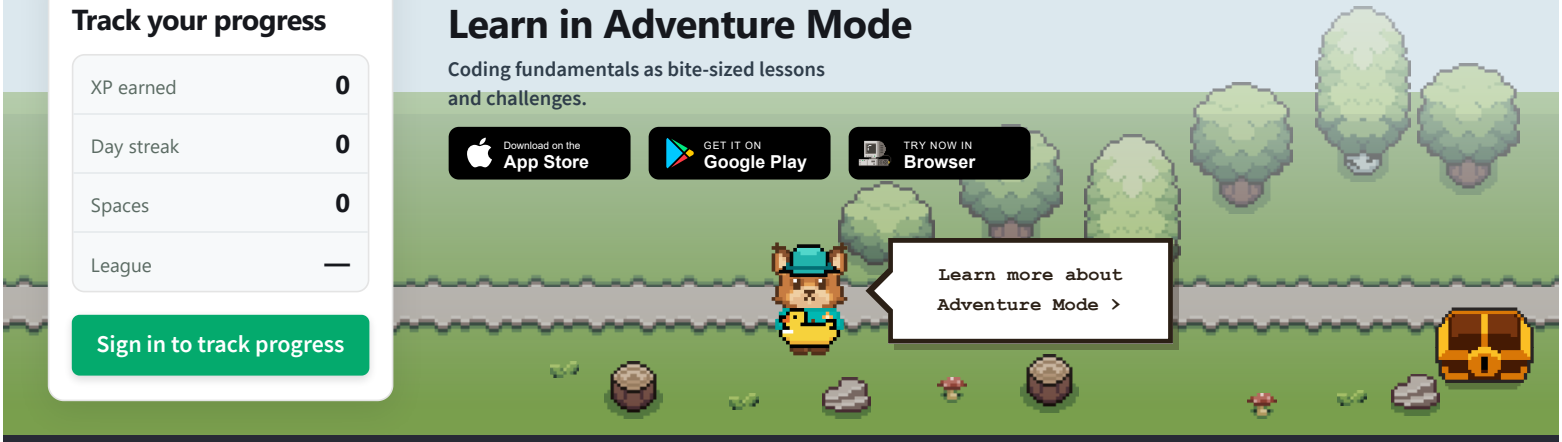


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JavaScript Array includes()

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Examples

```
const fruits = ["Banana", "Orange", "Apple", "Mango"];
fruits.includes("Mango");
```

Try it Yourself »

Start the search at position 3:

```
const fruits = ["Banana", "Orange", "Apple", "Mango"];
fruits.includes("Banana", 3);
```

Try it Yourself »

Description

SHARE

The `includes()` method returns `true` if an array contains a specified value.

The `includes()` method returns `false` if the value is not found.

The `includes()` method is case sensitive.

Array Find Methods:

Method	Finds
includes()	Returns true if an array contains a specified value
indexOf()	The index of the first element with a specified value
lastIndexOf()	The index of the last element with a specified value
find()	The value of the first element that passes a test
findIndex()	The index of the first element that passes a test
findLast()	The value of the last element that passes a test
findLastIndex()	The index of the last element that passes a test

Syntax

```
array.includes(element, start)
```

Parameters

Parameter	Description
element	Required. The value to search for.
start	Optional. Start position. Default is 0.

Return Value

Type	Description
A boolean	`true` if the value is found, otherwise `false`.

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Basic Array Methods

Array Search Methods

Array Sort Methods

Array Iteration Methods

Browser Support

`includes()` is a JavaScript 2016 feature.

ES 2016 is supported in all modern browsers since **March 2017**:

Chrome	Edge	Firefox	Safari	Opera
Yes	Yes	Yes	Yes	Yes

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JavaScript Array Reference

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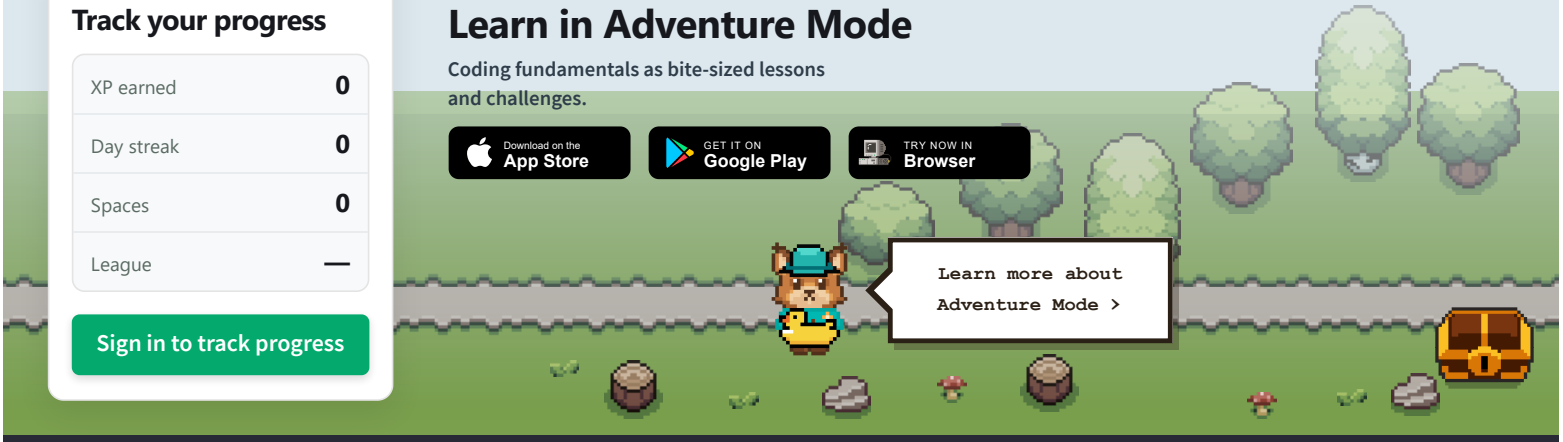
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JavaScript Array indexOf()

‹ Previous JavaScript Array Reference Next ›

Examples

Find the first index of "Apple":

```
const fruits = ["Banana", "Orange", "Apple", "Mango"];
let index = fruits.indexOf("Apple");
```

Try it Yourself »

Start at index 3:

```
const fruits = ["Banana", "Orange", "Apple", "Mango", "Apple"];
let index = fruits.indexOf("Apple", 3);
```

Try it Yourself »

More examples below.

Description

SHARE

The `indexOf()` method returns the first index (position) of a specified value.

The `indexOf()` method returns -1 if the value is not found.

The `indexOf()` method starts at a specified index and searches from left to right (from the given start postion to the end of the array).

By default the search starts at the first element and ends at the last.

Negative start values counts from the last element (but still searches from left to right).

Array Find Methods:

Method	Finds
includes()	Returns true if an array contains a specified value
indexOf()	The index of the first element with a specified value
lastIndexOf()	The index of the last element with a specified value
find()	The value of the first element that passes a test
findIndex()	The index of the first element that passes a test
findLast()	The value of the last element that passes a test
findLastIndex()	The index of the last element that passes a test

Syntax

```
array.indexOf(item, start)
```

Parameters

Parameter	Description
item	Required. The value to search for.
start	Optional. Where to start the search. Default value is 0. Negative values start the search from the end of the array.

Return Value

Type	Description

A number	The index (position) of the first item found. -1 if the item is not found.

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Note

In an array, the first element has index (position) 0, the second has index 1, ...

More Examples

Find the first index of "Apple", starting from the last element:

```
const fruits = ["Banana", "Orange", "Apple", "Mango", "Apple"];
let index = fruits.indexOf("Apple", -1);
```

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Basic Array Methods

Array Search Methods

Array Sort Methods

Array Iteration Methods

Browser Support

`indexOf()` is an ECMAScript5 (ES5 2009) feature.

JavaScript 2009 is supported in all browsers since **July 2013**:

Chrome	IE/Edge	Firefox	Safari	Opera
23	11	21	6	15
Sep 2012	Sep 2012	Apr 2013	Jul 2012	Jul 2013

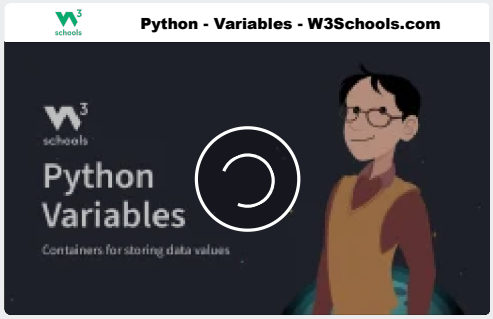
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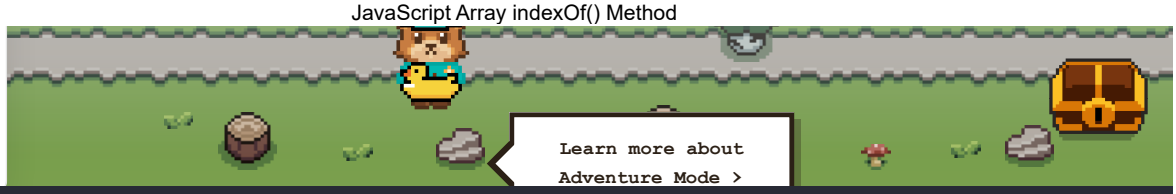
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JavaScript Array.isArray()

‹ Previous JavaScript Array Reference Next ›

Examples

Check if an object is an array:

```
const fruits = ["Banana", "Orange", "Apple", "Mango"];
let result = Array.isArray(fruits);
```

Try it Yourself »

Check if another datatype is an array:

```
let text = "W3Schools";
let result = Array.isArray(text);
```

Try it Yourself »

Description

SHARE

The `isArray()` method returns `true` if an object is an array, otherwise `false`.

Array.isArray()

Array.isArray() is a static property of the JavaScript Array object.

You can only use it as Array.isArray().

Using x.isArray(), where x is an array will return undefined.

Syntax

```
Array.isArray(obj)
```

Parameters

Parameter	Description
obj	Required. An object (or any data type) to be tested.

Return Value

Type	Description
A boolean	`true` if the object is an array, otherwise `false`.

Array Tutorials:

Array Tutorial

Array Const

Basic Array Methods

Array Search Methods

Array Sort Methods

Array Iteration Methods

Browser Support

`isArray()` is an ECMAScript5 (ES5 2009) feature.

JavaScript 2009 is supported in all browsers since **July 2013**:







Chrome 23	IE/Edge 11	Firefox 21	Safari 6	Opera 15
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JavaScript Array join()

‹ Previous JavaScript Array Reference Next ›

Examples

```
const fruits = ["Banana", "Orange", "Apple", "Mango"];
let text = fruits.join();
```

Try it Yourself »

Another separator:

```
const fruits = ["Banana", "Orange", "Apple", "Mango"];
let text = fruits.join(" and ");
```

Try it Yourself »

Description

SHARE

The `join()` method returns an array as a string.

The `join()` method does not change the original array.

Any separator can be specified. The default is comma (,).

Syntax

```
array.join(separator)
```

Parameters

Parameter	Description
separator	Optional. The separator to be used. Default is a comma.

Return Value

Type	Description
A string.	The array values, separated by the specified separator.

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Array Tutorials:

Array Tutorial

Array Const

Basic Array Methods

Array Search Methods

Array Sort Methods

Array Iteration Methods

Browser Support

`join()` is an ECMAScript1 (JavaScript 1997) feature.

It is supported in all browsers:

				
Chrome	Edge	Firefox	Safari	Opera

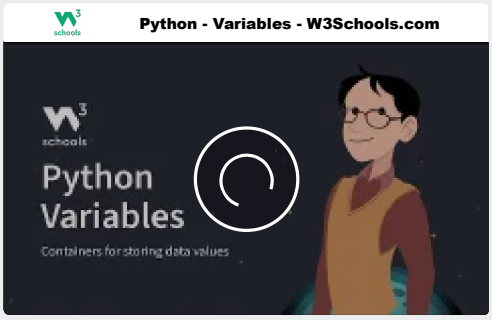
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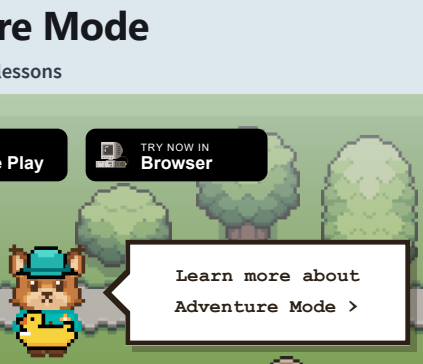
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JavaScript Array keys()

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Example

Create an Array Iterator object, containing the keys of the array:

```
// Create an Array
const fruits = ["Banana", "Orange", "Apple", "Mango"];

// Create an Iterable
const list = fruits.keys();

// List the Keys
let text = "";
for (let x of list) {
  text += x + "<br>";
}
```

Try it Yourself »

More Examples Below !

Description

SHARE

The `keys()` method returns an Iterator object with the keys of an array.

The `keys()` method does not change the original array.

Array Iteration Methods:

The Array entries() Method

The Array every() Method

The Array filter() Method

The Array forEach() Method

The Array keys() Method

The Array map() Method

Syntax

```
array.keys()
```

Parameters

NONE

Return Value

Type	Description
An array	An Array Iterator object containing the keys of an array.

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More Examples

Example

Iterate directly over the iterator:

```
// Create an Array
const fruits = ["Banana", "Orange", "Apple", "Mango"];

// List the Keys
```

```
let text = "";
for (let x of fruits.keys()) {
  text += x + "<br>";
}
```

Try it Yourself »

Example

Use the built in Object.keys() method:

```
// Create an Array
const fruits = ["Banana", "Orange", "Apple", "Mango"];

// List the Keys
let text = "";
for (let x of Object.keys(fruits)) {
  text += x + "<br>";
}
```

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Basic Array Methods

Array Search Methods

Array Sort Methods

Array Iteration Methods

Browser Support

`keys()` is an ECMAScript6 (ES6 2015) feature.

JavaScript 2015 is supported in all browsers since **June 2017**:

Chrome 51	Edge 15	Firefox 54	Safari 10	Opera 38
May 2016	Apr 2017	Jun 2017	Sep 2016	Jun 2016

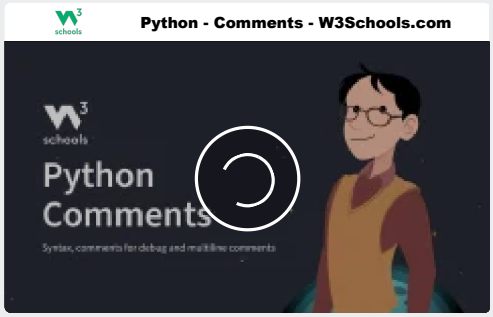
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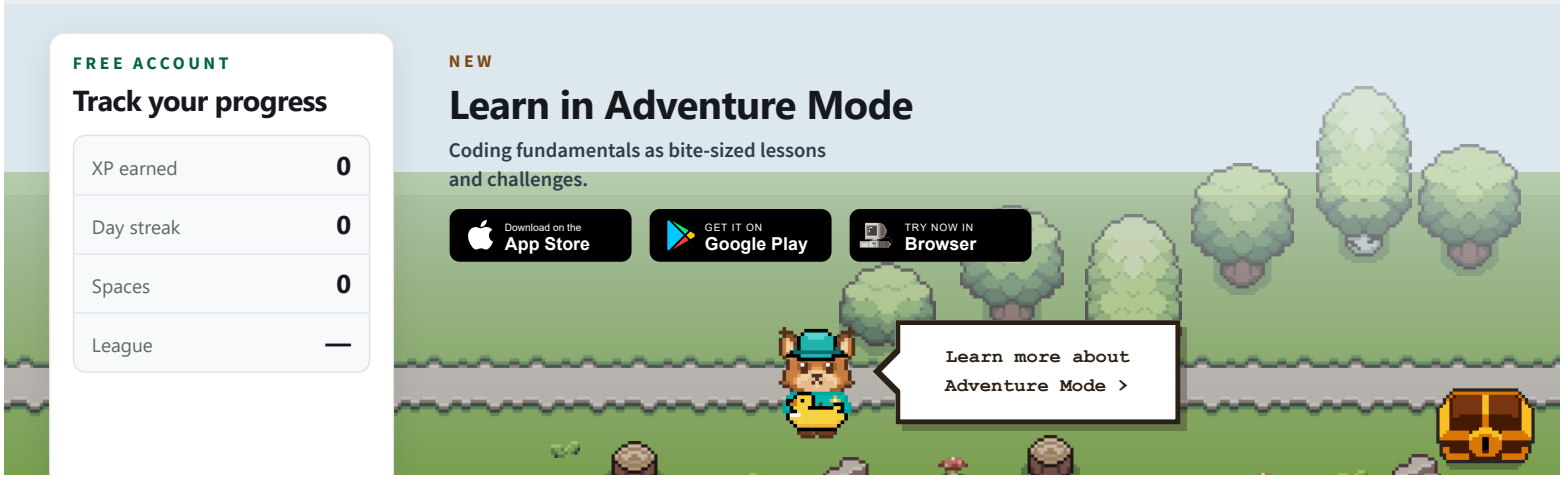
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JavaScript Array lastIndexOf()

‹ Previous JavaScript Array Reference Next ›

Examples

Find the last index of "Apple":

```
const fruits = ["Apple", "Orange", "Apple", "Mango"];
let index = fruits.lastIndexOf("Apple");
```

Try it Yourself »

More than one apple:

```
const fruits = ["Orange", "Apple", "Mango", "Apple", "Banana", "Apple"];
let index = fruits.lastIndexOf("Apple");
```

Try it Yourself »

More examples below.

Description

SHARE

The `lastIndexOf()` method returns the last index (position) of a specified value.

The `lastIndexOf()` method returns -1 if the value is not found.

The `lastIndexOf()` starts at a specified index and searches from right to left (from the given postion to the beginning of the array).

By defalt the search starts at the last element and ends at the first.

Negative start values counts from the last element (but still searches from right to left).

Array Find Methods:

Method	Finds
includes()	Returns true if an array contains a specified value
indexOf()	The index of the first element with a specified value
lastIndexOf()	The index of the last element with a specified value
find()	The value of the first element that passes a test
findIndex()	The index of the first element that passes a test
findLast()	The value of the last element that passes a test
findLastIndex()	The index of the last element that passes a test

Syntax

```
array.lastIndexOf(item, start)
```

Parameters

Parameter	Description
item	Required. The value to search for.
start	Optional. Where to start the search. Default is the last element (array.length-1). Negative start values counts from the last element (but still searches from right to left).

Return Value

Type	Description
A number	The position of the specified item. -1 if the item is not found.

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More Examples

Start the search at position 4:

```
const fruits = ["Orange", "Apple", "Mango", "Apple", "Banana", "Apple"];
let index = fruits.lastIndexOf("Apple", 4);
```

Try it Yourself »

Start the search at the second last position:

```
const fruits = ["Orange", "Apple", "Mango", "Apple", "Banana", "Apple"];
let index = fruits.lastIndexOf("Apple", -2);
```

Try it Yourself »

Array Tutorials:

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Basic Array Methods

Array Search Methods

Array Sort Methods

Array Iteration Methods

Browser Support

`lastIndexOf()` is an ECMAScript5 (ES5 2009) feature.

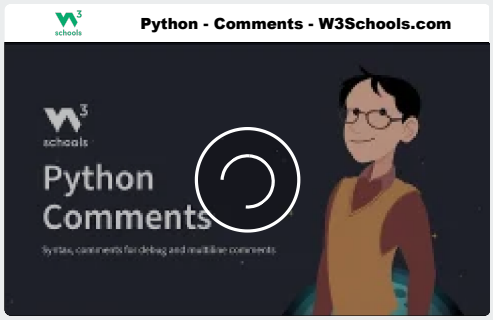
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Chrome	IE/Edge	Firefox	Safari	Opera
23	11	21	6	15
Sep 2012	Sep 2012	Apr 2013	Jul 2012	Jul 2013

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JavaScript Array length

‹ Previous JavaScript Array Reference Next ›

Examples

```
const fruits = ["Banana", "Orange", "Apple", "Mango"];
let length = fruits.length;
```

Try it Yourself »

```
const fruits = ["Banana", "Orange", "Apple", "Mango"];
fruits.length = 2;
```

Try it Yourself »

Description

SHARE

The `length` property sets or returns the number of elements in an array.

Syntax

Return the length of an array:

```
array.length
```

Set the length of an array:

```
array.length = number
```

Return Value

Type	Description
A number	The number of elements in the array.

Array Tutorials:

Array Tutorial

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Basic Array Methods

Array Search Methods

Array Sort Methods

Array Iteration Methods

Browser Support

`length` is an ECMAScript1 (JavaScript 1997) feature.

It is supported in all browsers:

				
Chrome	Edge	Firefox	Safari	Opera

‹ Previous JavaScript Array Reference Next ›

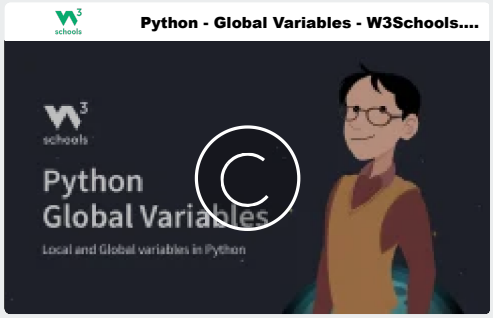
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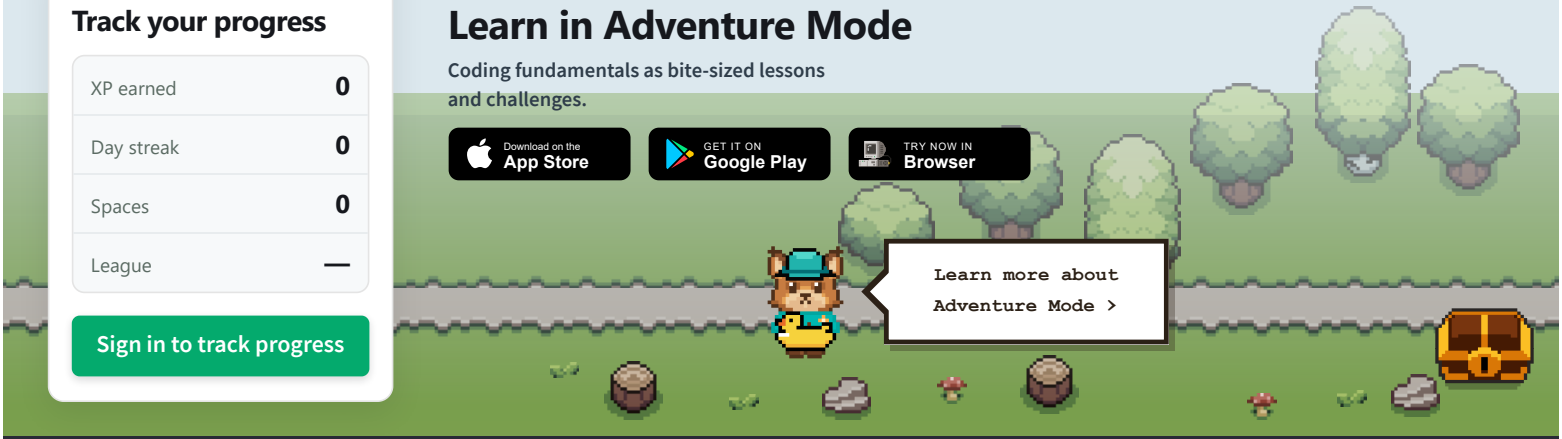
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JavaScript Array map()

‹ Previous JavaScript Array Reference Next ›

Examples

Return a new array with the square root of all element values:

```
const numbers = [4, 9, 16, 25];
const newArr = numbers.map(Math.sqrt)
```

Try it Yourself »

Multiply all the values in an array with 10:

```
const numbers = [65, 44, 12, 4];
const newArr = numbers.map(myFunction)

function myFunction(num) {
  return num * 10;
}
```

Try it Yourself »

More examples below.

Description

SHARE

`map()` creates a new array from calling a function for every array element.

`map()` does not execute the function for empty elements.

`map()` does not change the original array.

Array Iteration Methods:

The Array entries() Method

The Array every() Method

The Array filter() Method

The Array forEach() Method

The Array keys() Method

The Array map() Method

Syntax

```
array.map(function(currentValue, index, arr), thisValue)
```

Parameters

Parameter	Description
function()	Required. A function to be run for each array element.
currentValue	Required. The value of the current element.
index	Optional. The index of the current element.
arr	Optional. The array of the current element.
thisValue	Optional. Default value `undefined`. A value passed to the function to be used as its `this` value.

Return Value

Type	Description

An array	The results of a function for each array element.

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More Examples

Get the full name for each person:

```
const persons = [
  {firstname : "Malcom", lastname: "Reynolds"},
  {firstname : "Kaylee", lastname: "Frye"},
  {firstname : "Jayne", lastname: "Cobb"}
];

persons.map(getFullName);

function getFullName(item) {
  return [item.firstname,item.lastname].join(" ");
}
```

Try it Yourself »

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Basic Array Methods

Array Search Methods

Array Sort Methods

Array Iteration Methods

Browser Support

`map()` is an ECMAScript5 (ES5 2009) feature.

JavaScript 2009 is supported in all browsers since **July 2013**:

Chrome	IE/Edge	Firefox	Safari	Opera
23	11	21	6	15
Sep 2012	Sep 2012	Apr 2013	Jul 2012	Jul 2013

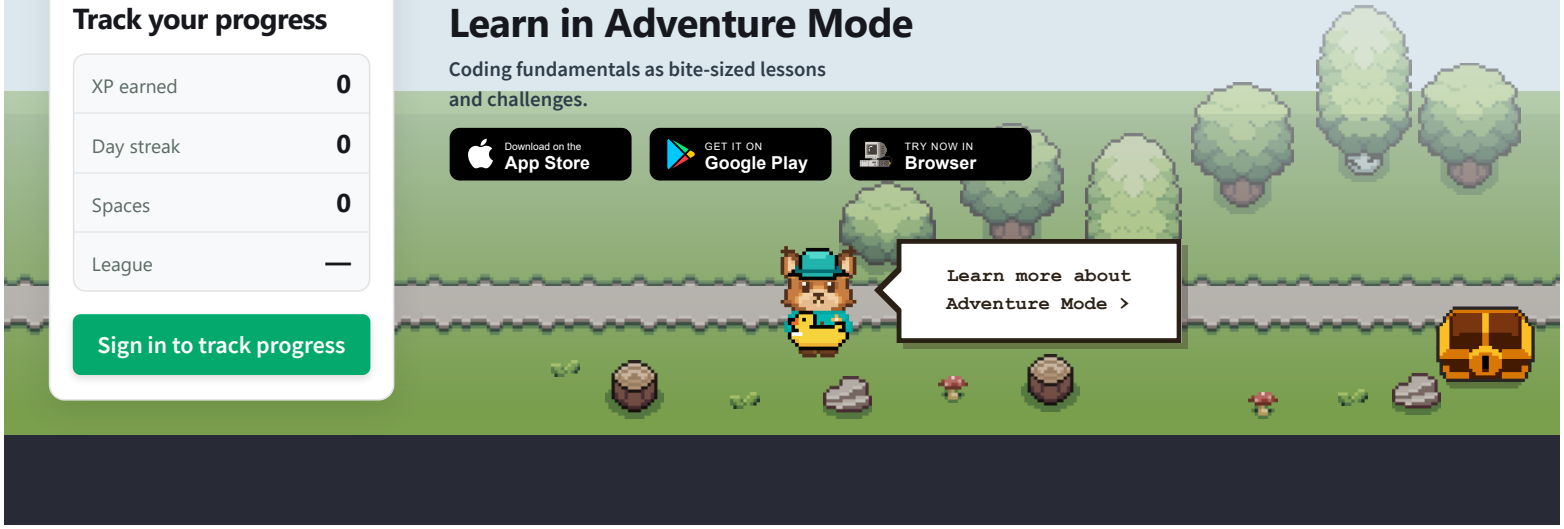
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JavaScript Array.of()

‹ Previous JavaScript Array Reference Next ›

Examples

Create a new array from a number of arguments:

```
let fruits = Array.of("Banana", "Orange", "Apple", "Mango");
document.getElementById("demo").innerHTML = fruits;
```

Try it Yourself »

Description

SHARE

The `Array.of()` method creates a new array from any number of arguments.

The `Array.of()` method can take any type of arguments.

Syntax

```
Array.of(element1, element2, ... , elementN)
```

Parameters

Parameter	Description

elements	Optional. Any number of elements regardless of type.

Return Value

Type	Description
An Array	A new array created from the arguments. The arguments can be strings, numbers, arrays, or any other allowed type.

Array Tutorials:

Array Tutorial

Array Const

Basic Array Methods

Array Search Methods

Array Sort Methods

Array Iteration Methods

Browser Support

`of()` is an ECMAScript6 (ES6 2015) feature.

JavaScript 2015 is supported in all browsers since **June 2017**:

				
Chrome 51	Edge 15	Firefox 54	Safari 10	Opera 38
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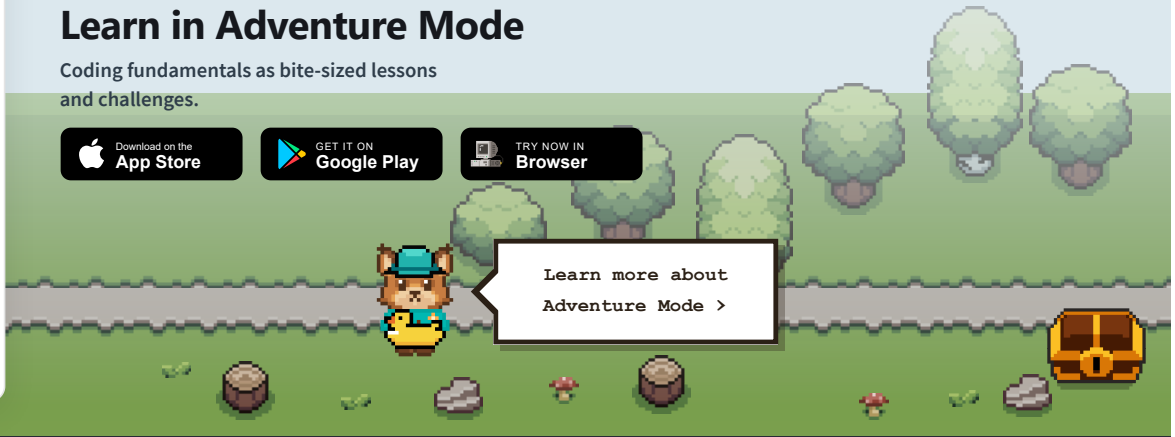
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JavaScript Array pop()

‹ Previous JavaScript Array Reference Next ›

Examples

Remove (pop) the last element:

```
const fruits = ["Banana", "Orange", "Apple", "Mango"];
fruits.pop();
```

Try it Yourself »

`pop()` returns the element it removed:

```
const fruits = ["Banana", "Orange", "Apple", "Mango"];
fruits.pop();
```

Try it Yourself »

Description

SHARE

The `pop()` method removes (pops) **the last element** of an array.

The `pop()` method changes the original array.

The `pop()` method returns the removed element.

See Also:

The Array push() Method

The Array shift() Method

The Array unshift() Method

Syntax

```
array.pop()
```

Parameters

NONE

Return Value

Type	Description
A variable	The removed item. A string, a number, an array, or any other type allowed in an array.

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Basic Array Methods

Array Search Methods

Array Sort Methods

Array Iteration Methods

Browser Support

`pop()` is an ECMAScript1 (JavaScript 1997) feature.

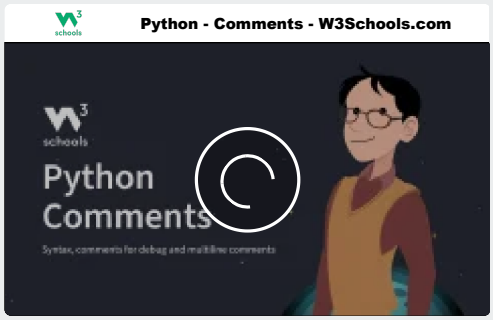
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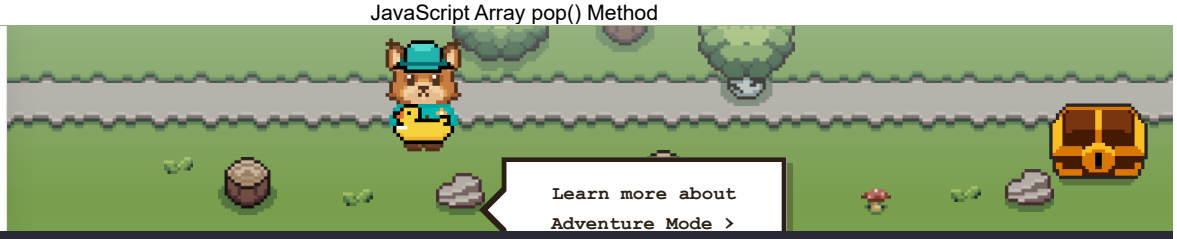
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JavaScript Array prototype

‹ Previous JavaScript Array Reference Next ›

Example

Create a method that transforms array values into upper case:

```
Array.prototype.myUcase = function() {
  for (let i = 0; i < this.length; i++) {
    this[i] = this[i].toUpperCase();
  }
};
```

Use the method on any array:

```
var fruits = ["Banana", "Orange", "Apple", "Mango"];
fruits.myUcase();
```

Try it Yourself »

Description

SHARE

`prototype` allows you to add new properties and methods to arrays.

`prototype` is a property available with all JavaScript objects.

Syntax

```
Array.prototype.name = value
```

Warning

You are not advised to change the prototype of an object that you do not control.

You should not change the prototype of built in JavaScript datatypes like:

- Numbers
- Strings
- Arrays
- Dates
- Booleans
- Function
- Objects

Only change the prototype of your own objects.

The prototype Property

The JavaScript `prototype` property allows you to add new properties to objects:

Example

```
function Person(first, last, age, eyecolor) {
  this.firstName = first;
  this.lastName = last;
  this.eyeColor = eyecolor;
}

Person.prototype.nationality = "English";
```

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Browser Support

`prototype` is an ECMAScript1 (JavaScript 1997) feature.

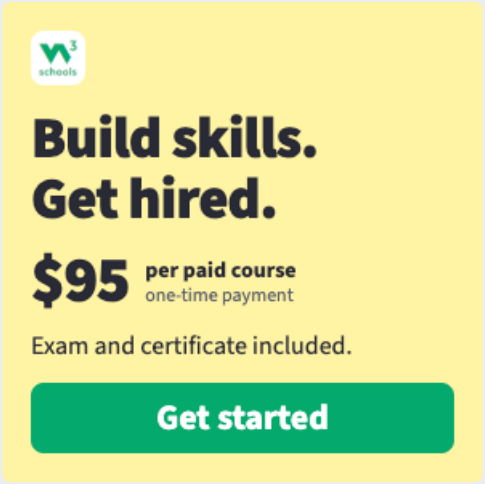
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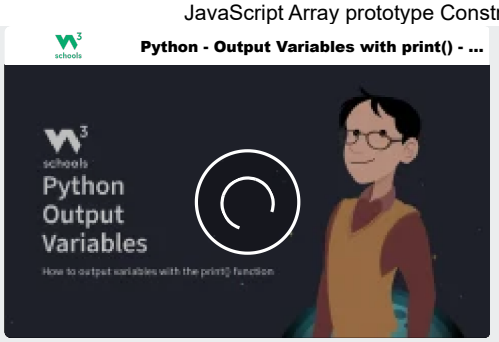
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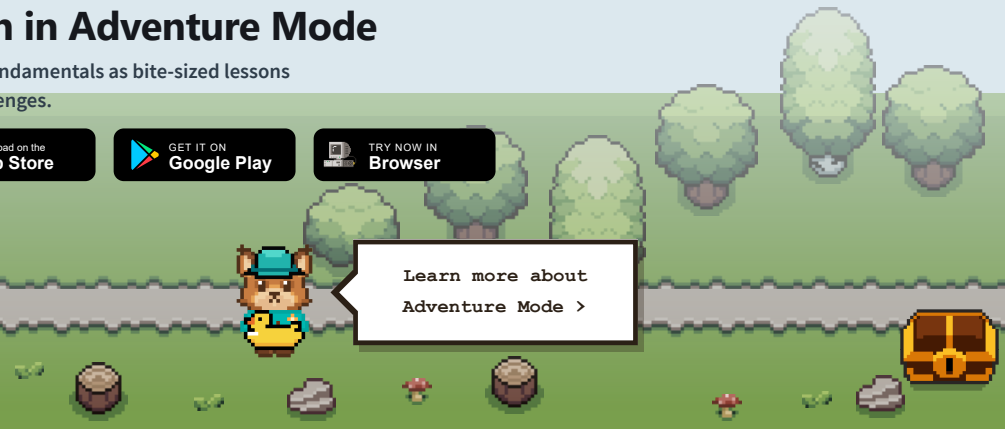
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JavaScript Array push()

‹ Previous JavaScript Array Reference Next ›

Examples

Add a new item to an array:

```
const fruits = ["Banana", "Orange", "Apple", "Mango"];
fruits.push("Kiwi");
```

Try it Yourself »

Add two new items to the array:

```
const fruits = ["Banana", "Orange", "Apple", "Mango"];
fruits.push("Kiwi", "Lemon");
```

Try it Yourself »

Description

SHARE

The `push()` method adds new items **to the end** of an array.

The `push()` method changes the length of the array.

The `push()` method returns the new length.

See Also:

The Array pop() Method

The Array shift() Method

The Array unshift() Method

Syntax

```
array.push(item1, item2, ..., itemX)
```

Parameters

Parameters	Description
item1 *item2* .. *itemX*	The item(s) to add to the array. Minimum one item is required.

Return Value

Type	Description
A number	The new length of the array.

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More Examples

Add 3 items to the array:

```
const fruits = ["Banana", "Orange", "Apple", "Mango"];
fruits.push("Kiwi", "Lemon", "Pineapple");
```

Try it Yourself »

`push()` returns the new length:

```
const fruits = ["Banana", "Orange", "Apple", "Mango"];
fruits.push("Kiwi");
```

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Array Tutorials:

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Basic Array Methods

Array Search Methods

Array Sort Methods

Array Iteration Methods

Browser Support

`push` is an ECMAScript1 (JavaScript 1997) feature.

It is supported in all browsers:

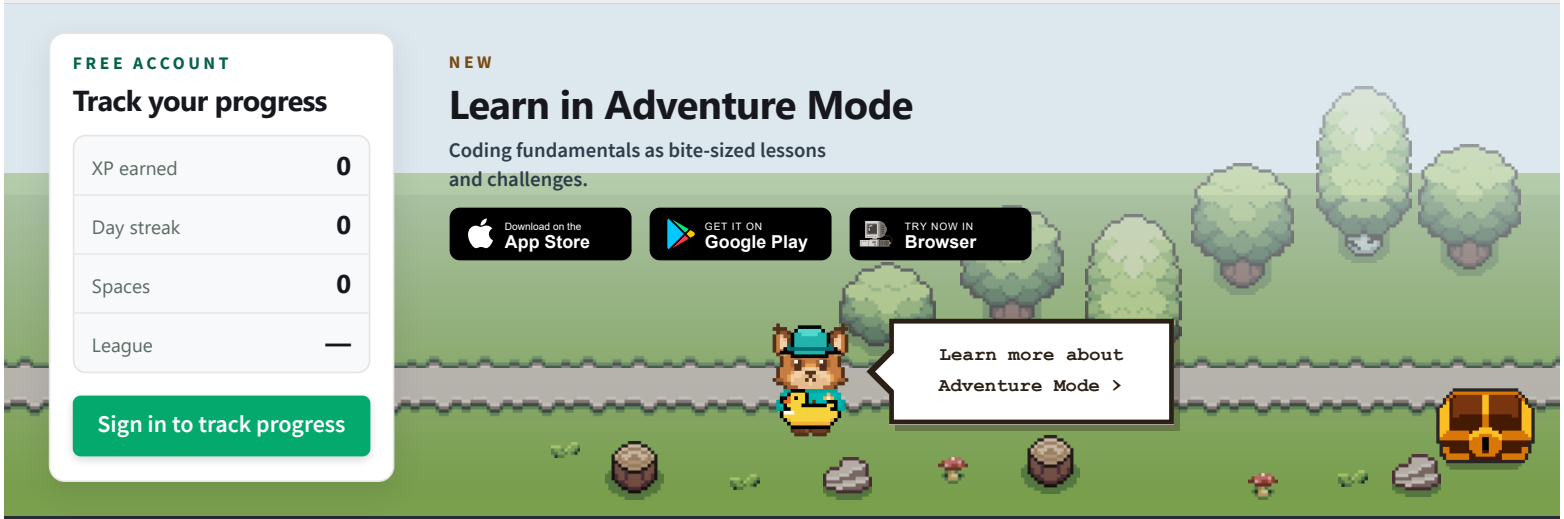
				
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JavaScript Array reduce()

‹ Previous JavaScript Array Reference Next ›

Examples

Subtract all numbers in an array:

```
const numbers = [175, 50, 25];

document.getElementById("demo").innerHTML = numbers.reduce(myFunc);

function myFunc(total, num) {
  return total - num;
}
```

Try it Yourself »

Round all the numbers and display the sum:

```
const numbers = [15.5, 2.3, 1.1, 4.7];
document.getElementById("demo").innerHTML = numbers.reduce(getSum, 0);

function getSum(total, num) {
  return total + Math.round(num);
}
```

Try it Yourself »

Description

SHARE

The `reduce()` method executes a reducer function for array element.

The `reduce()` method returns a single value: the function's accumulated result.

The `reduce()` method does not execute the function for empty array elements.

The `reduce()` method does not change the original array.

Note

At the first callback, there is no return value from the previous callback.

Normally, array element 0 is used as initial value, and the iteration starts from array element 1.

If an initial value is supplied, this is used, and the iteration starts from array element 0.

See Also:

The Array reduceRight() Method

Syntax

```
array.reduce(function(total, currentValue, currentIndex, arr), initialValue)
```

Parameters

Parameter	Description
function()	Required. A function to be run for each element in the array.
Reducer function parameters:	
total	Required. The *initialValue*, or the previously returned value of the function.
currentValue	Required. The value of the current element.
currentIndex	Optional. The index of the current element.

arr	Optional. The array the current element belongs to.

initialValue	Optional. A value to be passed to the function as the initial value.

Return Value

The accumulated result from the last call of the callback function.

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Array Search Methods

Array Sort Methods

Array Iteration Methods

Browser Support

`reduce()` is an ECMAScript5 (ES5 2009) feature.

JavaScript 2009 is supported in all browsers since **July 2013**:

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23	11	21	6	15
Sep 2012	Sep 2012	Apr 2013	Jul 2012	Jul 2013

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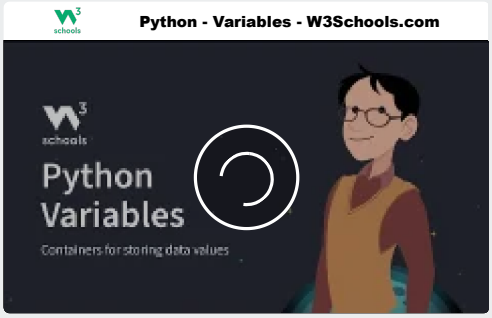
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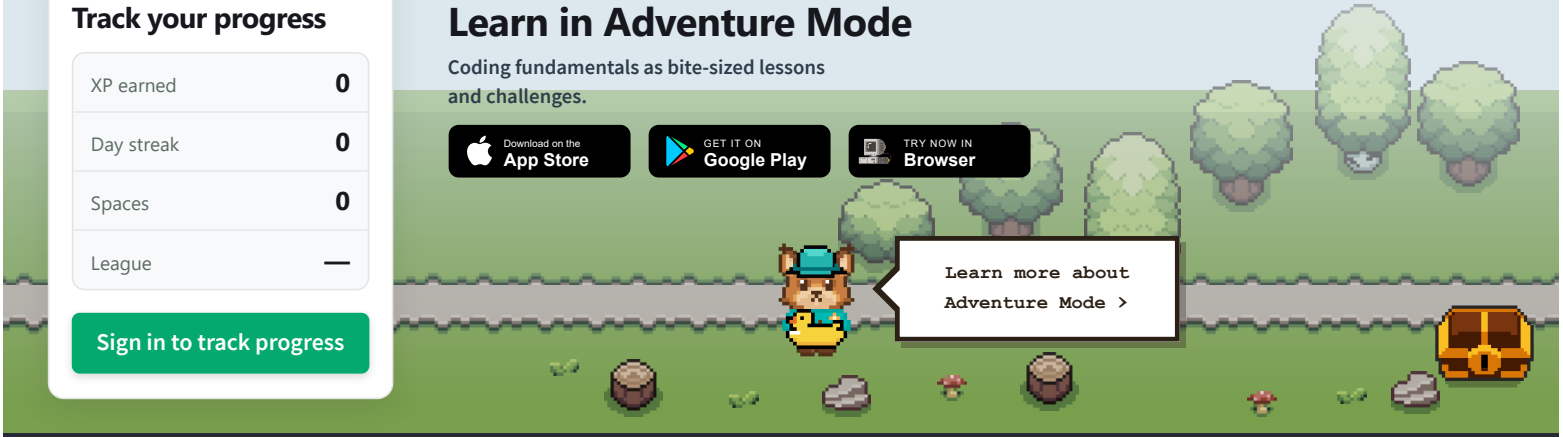
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JavaScript Array reduceRight()

‹ Previous JavaScript Array Reference Next ›

Examples

Subtract the numbers in the array, starting from the end:

```
const numbers = [175, 50, 25];

document.getElementById("demo").innerHTML = numbers.reduceRight(myFunc);

function myFunc(total, num) {
  return total - num;
}
```

Try it Yourself »

Subtract the numbers, right-to-left, and display the sum:

```
const numbers = [2, 45, 30, 100];
document.getElementById("demo").innerHTML = numbers.reduceRight(getSum);

function getSum(total, num) {
  return total - num;
}
```

Try it Yourself »

Description

SHARE

The `reduceRight()` method executes a reducer function for each array element.

The `reduceRight()` method works from right to left.

The `reduceRight()` method returns a single value: the function's accumulated result.

The `reduceRight()` method does not execute the function for empty elements.

Note

At the first callback, there is no return value from the previous callback.

Normally, the last array element is used as initial value, and the iteration starts from the element before.

If an initial value is supplied, this is used, and the iteration starts from last element.

See Also:

The Array reduce() Method

Syntax

```
array.reduceRight(function(total, currentValue, currentIndex, arr), initialValue)
```

Parameters

Parameter	Description
function()	Required. A function to be run for each element in the array.

Reducer function parameters:

total	Required. The *initialValue*, or the previously returned value of the function.
currentValue	Required. The value of the current element.
currentIndex	Optional. The index of the current element.

arr	Optional. The array the element belongs to.

initialValue	Optional. A value to be passed to the function as the initial value

Return Value

The accumulated result from the last call of the callback function.

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Browser Support

`reduceRight()` is an ECMAScript5 (ES5 2009) feature.

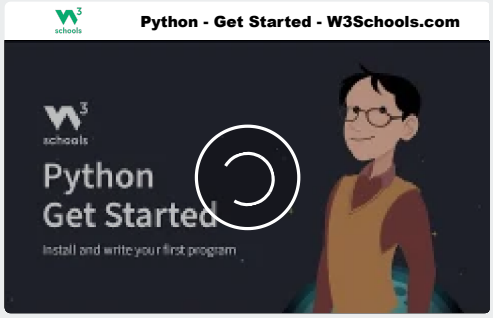
JavaScript 2009 is supported in all browsers since **July 2013**:

				
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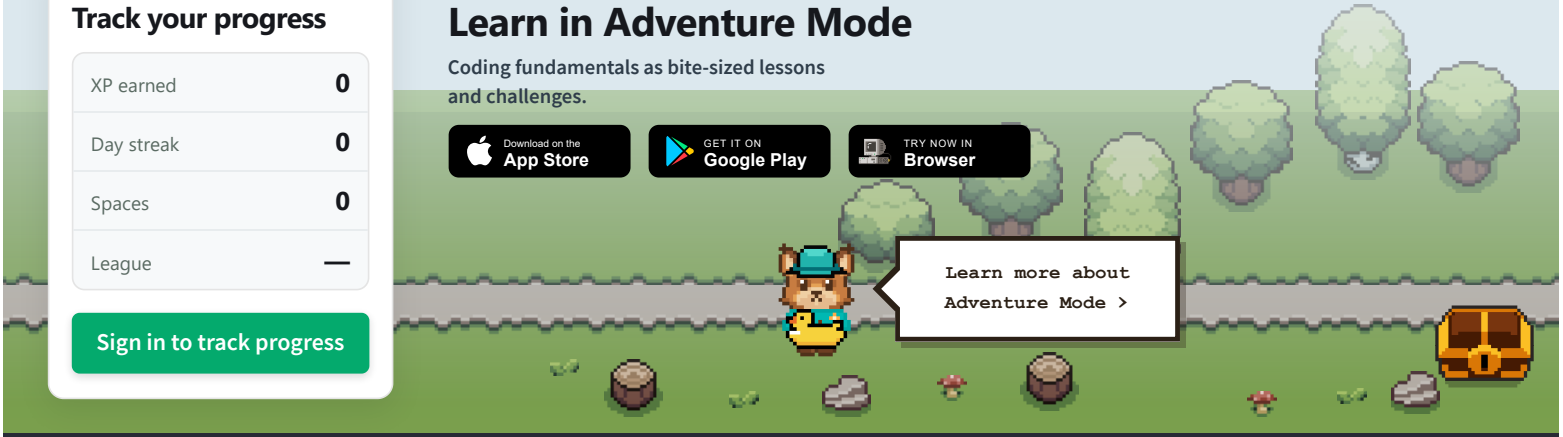
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JavaScript Array reverse()

‹ Previous JavaScript Array Reference Next ›

Example

```
const fruits = ["Banana", "Orange", "Apple", "Mango"];
fruits.reverse();
```

Try it Yourself »

Description

SHARE

The `reverse()` method reverses the order of the elements in an array.

The `reverse()` method overwrites the original array.

Array Sort Methods:

Method	Finds
reverse()	Reverses the order of the elements in an array
sort()	Sorts the elements of an array
toReversed()	Reverses the elements of an array into a new array

toSorted()	Sorts the elements of an array into a new array

Syntax

```
array.reverse()
```

Parameters

NONE

Return Value

Type	Description
Array	The array after it has been reversed.

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Array Search Methods

Array Sort Methods

Array Iteration Methods

Browser Support

`reverse()` is an ECMAScript1 (JavaScript 1997) feature.

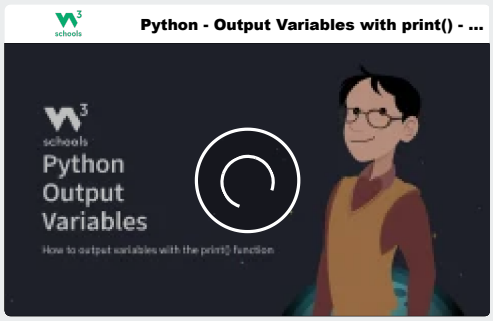
It is supported in all browsers:

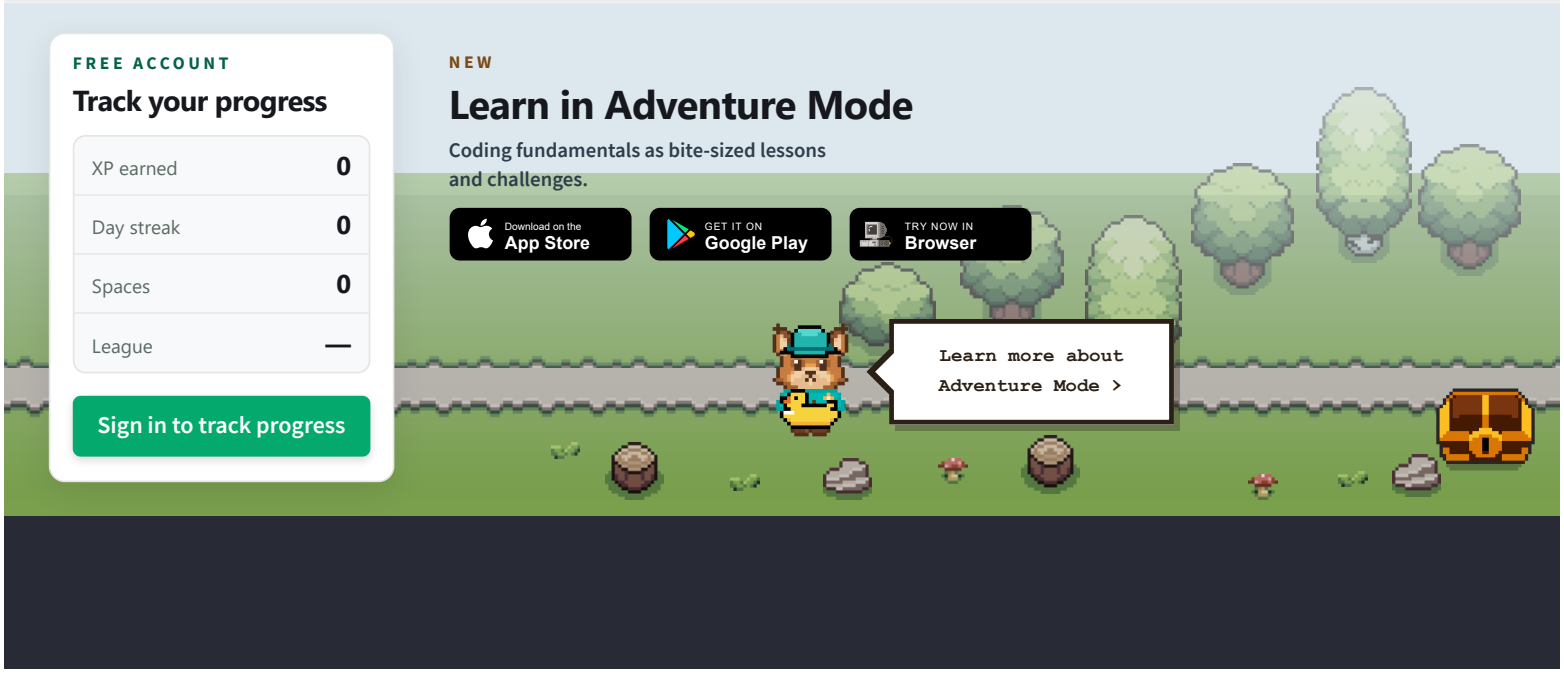
				
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JavaScript Array shift()

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Examples

Shift (remove) the first element of the array:

```
const fruits = ["Banana", "Orange", "Apple", "Mango"];
fruits.shift();
```

Try it Yourself »

The `shift()` method returns the shifted element:

```
const fruits = ["Banana", "Orange", "Apple", "Mango"];
fruits.shift();
```

Try it Yourself »

Description

SHARE

The `shift()` method removes **the first item** of an array.

The `shift()` method changes the original array.

The `shift()` method returns the shifted element.

See Also:

The Array unshift() Method

The Array push() Method

The Array pop() Method

Syntax

```
array.shift()
```

Parameters

NONE

Return Value

Type	Description
A variable	The removed item. A string, a number, an array, or any other type allowed in an array.

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Array Iteration Methods

Browser Support

`shift()` is an ECMAScript1 (JavaScript 1997) feature.

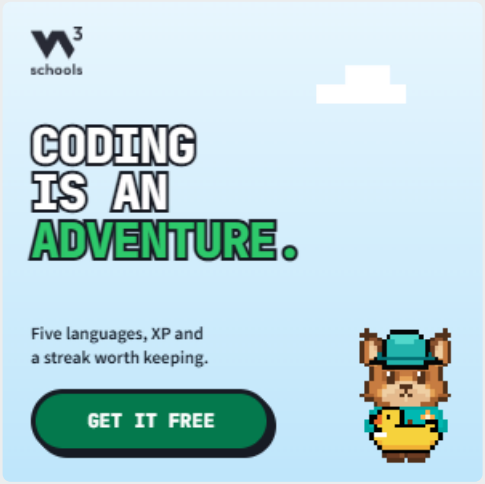
It is supported in all browsers:

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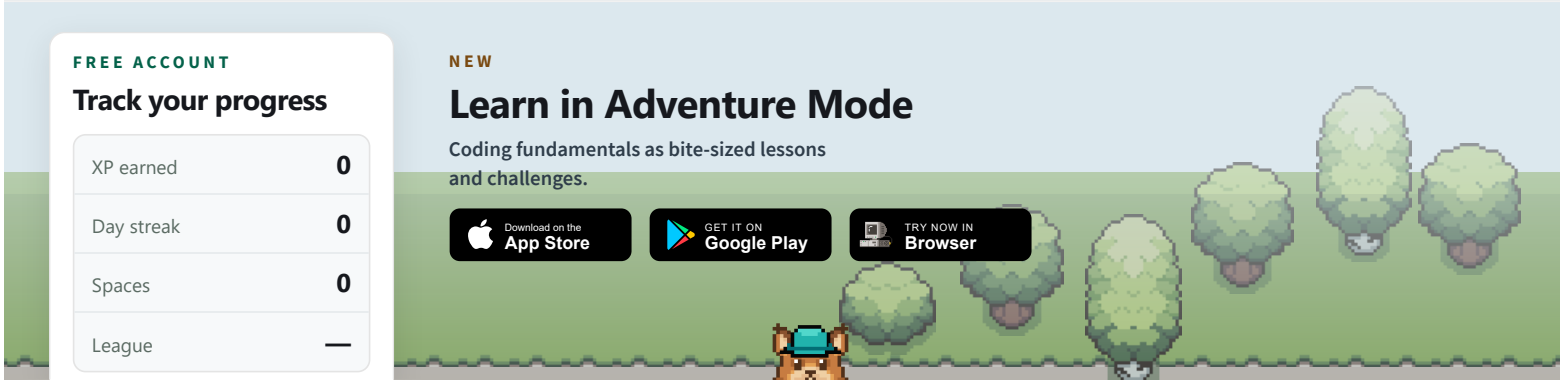
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JavaScript Array slice()

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Examples

Select elements:

```
const fruits = ["Banana", "Orange", "Lemon", "Apple", "Mango"];
const citrus = fruits.slice(1, 3);
```

Try it Yourself »

Select elements using negative values:

```
const fruits = ["Banana", "Orange", "Lemon", "Apple", "Mango"];
const myBest = fruits.slice(-3, -1);
```

Try it Yourself »

Description

SHARE

The `slice()` method returns selected elements in a new array.

The `slice()` method selects from a given *start*, up to a (not inclusive) given *end*.

The `slice()` method does not change the original array.

See Also:

The Array splice() Method

The Array toSpliced() Method

Syntax

```
array.slice(start, end)
```

Parameters

Parameter	Description
start	Optional. Start position. Default is 0. Negative numbers select from the end of the array.
end	Optional. End position. Default is last element. Negative numbers select from the end of the array.

Return Value

A new array containing the selected elements.

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Browser Support

`slice()` is an ECMAScript1 (JavaScript 1997) feature.

It is supported in all browsers:

Chrome	Edge	Firefox	Safari	Opera

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JavaScript Array Reference

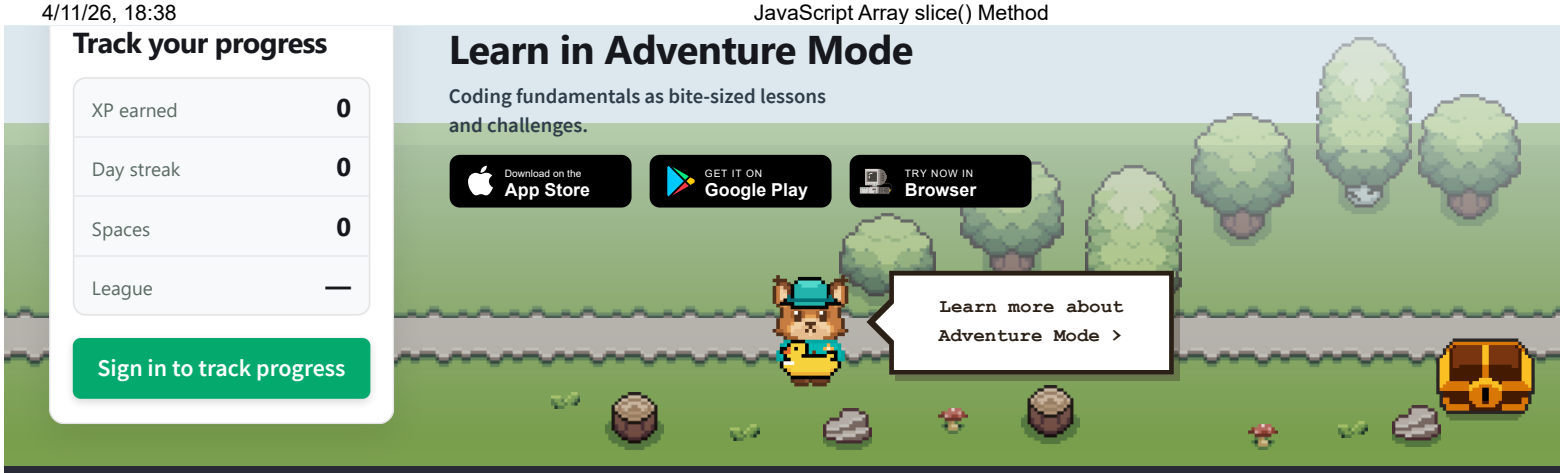
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JavaScript Array some()

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Example 1

Check if any values are over 18:

```
const ages = [3, 10, 18, 20];

ages.some(checkAdult);
function checkAdult(age) {
  return age > 18;
}
```

Try it Yourself »

Description

SHARE

The `some()` method checks if any array elements pass a test (provided as a callback function).

The `some()` method executes the callback function once for each array element.

The `some()` method returns `true` (and stops) if the function returns `true` for one of the array elements.

The `some()` method returns `false` if the function returns `false` for all of the array elements.

The `some()` method does not execute the function for empty array elements.

The `some()` method does not change the original array.

Syntax

```
array.some(function(value, index, arr), this)
```

Parameters

Parameter	Description
function	Required. A function to run for each array element.
Function parameters:	
value	Required. The value of the current element.
index	Optional. The index of the current element.
arr	Optional. The array the current element belongs to.
this	Optional. Default undefined. A value passed to the function to be used as its "this" value.

Return Value

Type	Description
A boolean	`true` if any of the array elements pass the test, otherwise `false`.

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Example 2

```
<p>Input: <input type="number" id="toCheck" value="15"></p>

<button onclick="myFunction()">Test</button>

<p>Values higher: <span id="demo"></span></p>

<script>
```

```
const numbers = [4, 12, 16, 20];

function checkValue(x) {
  return x > document.getElementById("toCheck").value;
}

function myFunction() {
  document.getElementById("demo").innerHTML = numbers.some(checkValue);
}
</script>
```

Try it Yourself »

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Browser Support

`some()` is an ECMAScript3 (JavaScript 1999) feature.

It is supported in all browsers:

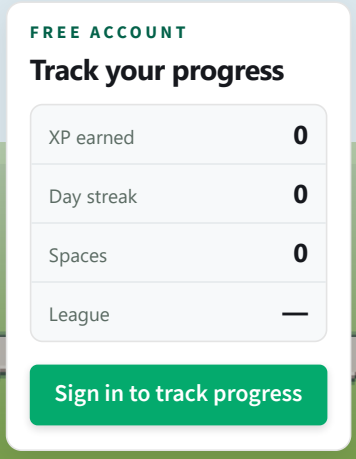
				
Chrome	Edge	Firefox	Safari	Opera

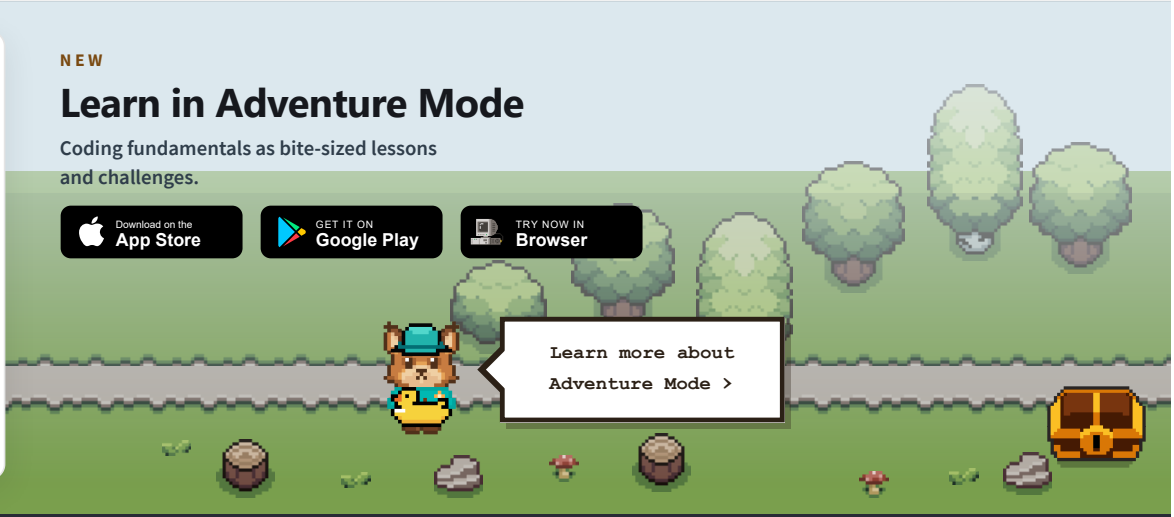
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JavaScript Array sort()

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Examples

```
// Create an Array
const fruits = ["Banana", "Orange", "Apple", "Mango"];

// Sort the Array
fruits.sort();
```

Try it Yourself »

More Examples Below !

Description

SHARE

The `sort()` method sorts the elements of an array.

The `sort()` method sorts the elements as strings in alphabetical and ascending order.

The `sort()` method overwrites the original array.

Array Sort Methods:

Method	Finds

reverse()	Reverses the order of the elements in an array
sort()	Sorts the elements of an array
toReversed()	Reverses the elements of an array into a new array
toSorted()	Sorts the elements of an array into a new array

Sort Compare Function

Sorting alphabetically works well for strings ("Apple" comes before "Banana").

But, sorting numbers can produce incorrect results.

"25" is bigger than "100", because "2" is bigger than "1".

You can fix this by providing a "compare function" (See examples below).

Syntax

```
array.sort(compareFunction)
```

Parameters

Parameter	Description
compareFunction	Optional. A function that defines a sort order. The function should return a negative, zero, or positive value, depending on the arguments: • function(a, b){return a-b} When sort() compares two values, it sends the values to the compare function, and sorts the values according to the returned (negative, zero, positive) value. **Example:** The sort function will sort 40 as a value lower than 100. When comparing 40 and 100, sort() calls the function(40,100). The function calculates 40-100, and returns -60 (a negative value).

Return Value

Type	Description
Array	The array with the items sorted.

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More Examples

Sort Decending

Sort and then reverse the order:

```
// Create an Array
const fruits = ["Banana", "Orange", "Apple", "Mango"];

// Sort the Array
fruits.sort();

// Reverse the array
fruits.reverse();
```

Try it Yourself »

Numeric Sorts

Using a Sort Function

Sort numbers in ascending order:

```
// Create an Array
const points = [40, 100, 1, 5, 25, 10];

// Sort the Array
points.sort(function(a, b){return a-b});
```

Try it Yourself »

Sort numbers in descending order:

```
// Create an Array
const points = [40, 100, 1, 5, 25, 10];

// Sort the Array
points.sort(function(a, b){return b-a});
```

Try it Yourself »

Find the lowest value:

```
// Create an Array
const points = [40, 100, 1, 5, 25, 10];

// Sort the numbers in ascending order
points.sort(function(a, b){return a-b});

let lowest = points[0];
```

Try it Yourself »

Find the highest value:

```
// Create an Array
const points = [40, 100, 1, 5, 25, 10];

// Sort the numbers in descending order:
points.sort(function(a, b){return b-a});

let highest = points[0];
```

Try it Yourself »

Find the highest value:

```
// Create an Array
const points = [40, 100, 1, 5, 25, 10];

// Sort the numbers in ascending order:
points.sort(function(a, b){return a-b});

let highest = points[points.length-1];
```

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Basic Array Methods

Array Search Methods

Array Sort Methods

Array Iteration Methods

Browser Support

`sort()` is an ECMAScript1 (JavaScript 1997) feature.

It is supported in all browsers:

Chrome	Edge	Firefox	Safari	Opera

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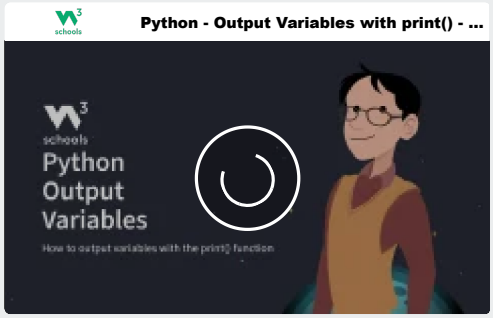
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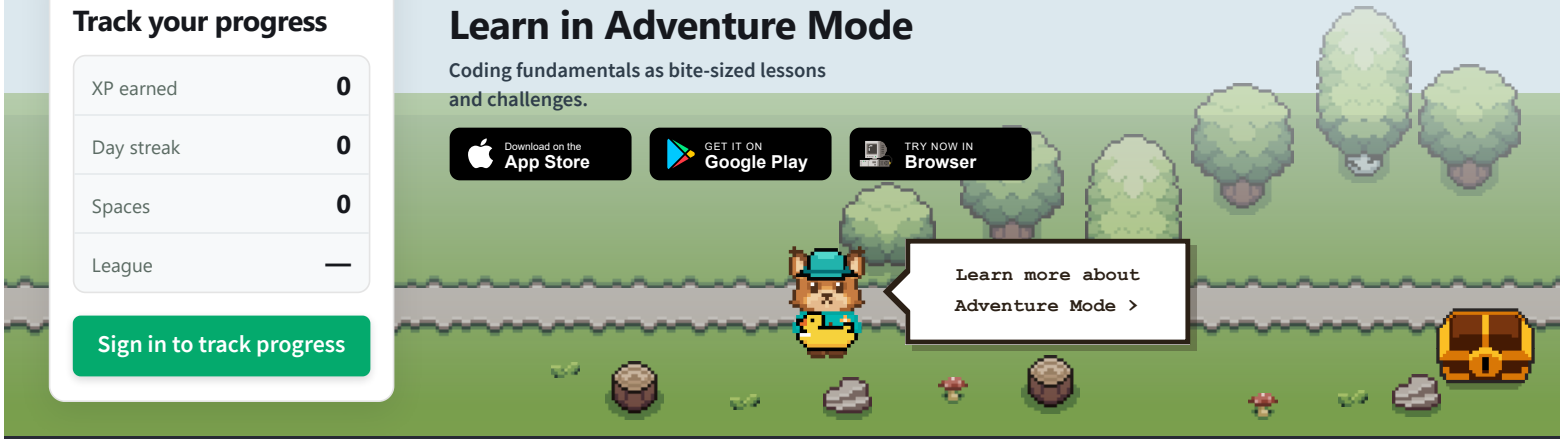
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JavaScript Array splice()

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Examples

```
// Create an Array
const fruits = ["Banana", "Orange", "Apple", "Mango"];

// At position 2, add "Lemon" and "Kiwi":
fruits.splice(2, 0, "Lemon", "Kiwi");
```

Try it Yourself »

More Examples Below !

Description

SHARE

The `splice()` method adds and/or removes array elements.

The `splice()` method overwrites the original array.

See Also:

The Array toSpliced() Method

The Array slice() Method

Syntax

```
array.splice(index, count, item1, ....., itemX)
```

Parameters

Parameter	Description
index	Required. The index (position) to add or remove items. A negative value counts from the end of the array.
count	Optional. Number of items to be removed.
item1, ...,	Optional. The new elements(s) to be added.

Return Value

Type	Description
Array	An array containing the removed items (if any).

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More Examples

Example

```
// Create an Array
const fruits = ["Banana", "Orange", "Apple", "Mango"];

// At position 2, remove 2 items
fruits.splice(2, 2);
```

Try it Yourself »

Example

```
// Create an Array
const fruits = ["Banana", "Orange", "Apple", "Mango"];

// At position 2, remove 1 item, add "Lemon" and "Kiwi"
fruits.splice(2, 1, "Lemon", "Kiwi");
```

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Browser Support

`splice()` is an ECMAScript1 (JavaScript 1997) feature.

It is supported in all browsers:

				
Chrome	Edge	Firefox	Safari	Opera

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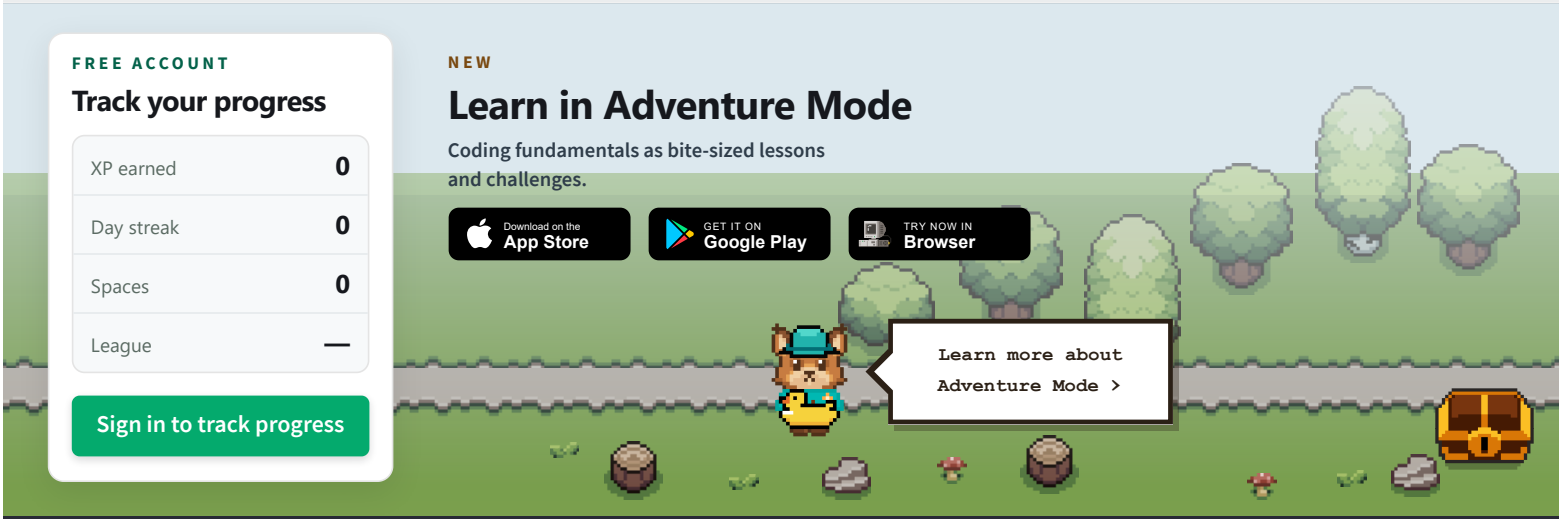
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JavaScript Operator Precedence

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Precedence describes the order in which operations are performed in an arithmetic expression.

As in traditional mathematics, multiplication is done first:

```
let x = 100 + 50 * 3;
```

Try it Yourself »

When using parentheses, operations inside the parentheses are computed first:

```
let x = (100 + 50) * 3;
```

Try it Yourself »

When operators have the same precedence (like + and -), they are computed from left to right:

```
let x = 100 / 50 * 3;
```

Try it Yourself »

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Operator Precedence Values

Expressions in parentheses are computed **before** the rest of the expression
Functions are executed **before** the result is used in the rest of the expression

Val	Operator	Description	Example
18	( )	Expression Grouping	(100 + 50) * 3
17	.	Member Of	car.name
17	[]	Member Of	car["name"]
17	?.	Optional Chaining ES2020	x ?. y
17	()	Function Call	myFunction()
17	new	New with Arguments	new Date("June 5,2022")
16	new	New without Arguments	new Date()
		Increment Operators Postfix increments are executed **before** prefix increments	
15	++	Postfix Increment	i++
15	--	Postfix Decrement	i--
14	++	Prefix Increment	++i
14	--	Prefix Decrement	--i
		NOT Operators	
14	!	Logical NOT	!(x==y)
14	~	Bitwise NOT	~x
		Unary Operators	
14	+	Unary Plus	+x
14	-	Unary Minus	-x
14	typeof	Data Type	typeof x
14	void	Evaluate Void	void(0)
14	delete	Property Delete	delete myCar.color
		Arithmetic Operators Exponentiations are executed **before** multiplications Multiplications and divisions are executed **before** additions and subtractions	

13	**	Exponentiation ES2016	10 ** 2
12	*	Multiplication	10 * 5
12	/	Division	10 / 5
12	%	Division Remainder	10 % 5
11	+	Addition	10 + 5
11	-	Subtraction	10 - 5
11	+	Concatenation	"John" + "Doe"
		Shift Operators	
10	<<	Shift Left	x << 2
10	>>	Shift Right (signed)	x >> 2
10	>>>	Shift Right (unsigned)	x >>> 2
		Relational Operators	
9	in	Property in Object	"PI" in Math
9	instanceof	Instance of Object	x instanceof Array
		Comparison Operators	
9	<	Less than	x < y
9	<=	Less than or equal	x <= y
9	>	Greater than	x > y
9	>=	Greater than or equal	x >= Array
8	==	Equal	x == y
8	===	Strict equal	x === y
8	!=	Unequal	x != y
8	!==	Strict unequal	x !== y
		Bitwise Operators	
7	&	Bitwise AND	x & y
6	^	Bitwise XOR	x ^ y
5	|	Bitwise OR	x | y
		Logical Operators	

4	&&	Logical AND	x && y
3	||	Logical OR	x || y
3	??	Nullish Coalescing ES2020	x ?? y
		Conditional (ternary) Operator	
2	? :	Condition	? "yes" : "no"
		Assignment Operators Assignments are executed **after** other operations	
2	=	Simple Assignment	x = y
2	+=	Addition Assignment	x += y
2	-=	Subtraction Assignment	x -= y
2	*=	Multiplication Assignment	x *= y
2	**=	Exponentiation Assignment	x **= y
2	/=	Division Assignment	x /= y
2	%=	Remainder Assignment	x %= y
2	<<=	Left Shift Assignment	x <<= y
2	>>=	Right Shift Assignment	x >>= y
2	>>>=	Unsigned Right Shift	x >>>= y
2	&=	Bitwise AND Assignment	x &= y
2	|=	Bitwise OR Assignment	x |= y
2	^=	Bitwise XOR Assignment	x ^= y
2	&&=	Logical AND Assignment	x &= y
2	||=	Logical OR Assignment	x ||= y
2	:	Colon Assignment	x : 5
2	=>	Arrow	x => y
2	yield	Pause / Resume	yield x
2	yield*	Delegate	yield* x
2	...	Spread	...x
1	,	Comma	x , y

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JavaScript Array []

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Example

```
// Create an Array
const cars = ["Saab", "Volvo", "BMW"];
```

Try it Yourself »

More Examples Below !

Description

The `[ ]` constructor creates a new array object.

Syntax

```
[iterable]
```

Parameters

Parameter	Description

iterable	Required. An iterable object with values.

Return Value

Type	Description
Array	A new Array object.

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More Examples

Example

Create an empty array and add values:

```
// Create an Array
const cars = [];

// Add Values to the Set
cars.push("Saab");
cars.push("Volvo");
cars.push("BMW");
```

Try it Yourself »

Example

Create an array with the new Array() method:

```
// Create an Array
const cars = new Array(["Saab", "Volvo", "BMW"]);
```

Try it Yourself »

Browser Support

`the [] array constructor` is an ECMAScript1 (JavaScript 1997) feature.

It is supported in all browsers:

Chrome	Edge	Firefox	Safari	Opera

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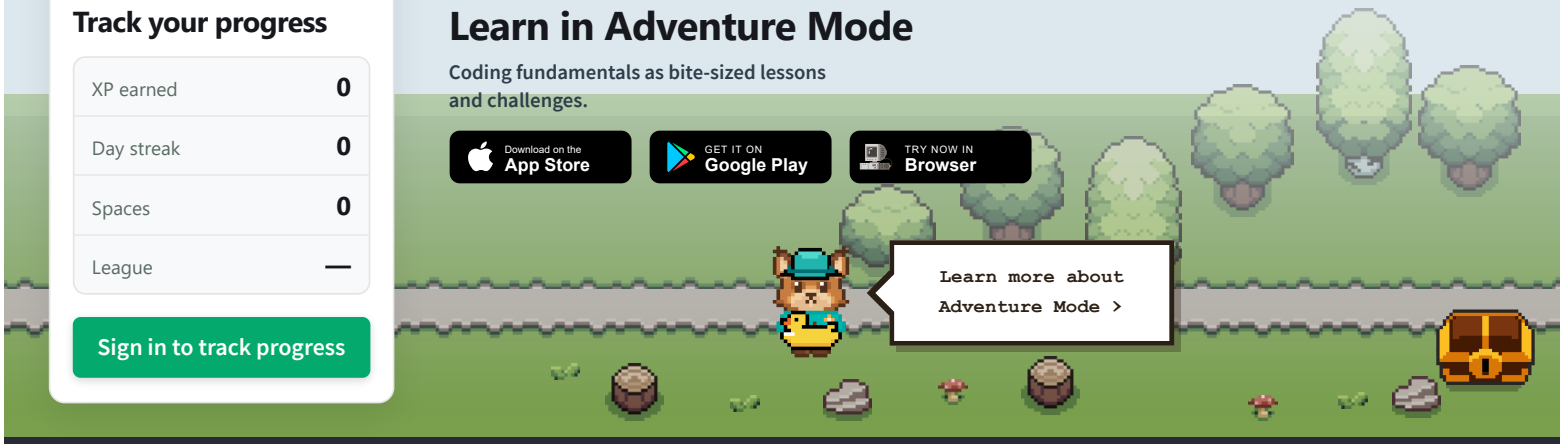
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JavaScript Array toReversed()

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Example

```
// Create an Array
const fruits = ["Banana", "Orange", "Apple", "Mango"];

// Reverse the Array
const fruits2 = fruits.toReversed();
```

Try it Yourself »

Description

SHARE

The `toReversed()` method reverses the order of the elements in an array.

The `toReversed()` method returns a new array.

The `toReversed()` method **does not** overwrites the original array.

The `toReversed()` method is the **copying version** of the `reverse()` method.

Array Sort Methods:

Method	Finds

reverse()	Reverses the order of the elements in an array
sort()	Sorts the elements of an array
toReversed()	Reverses the elements of an array into a new array
toSorted()	Sorts the elements of an array into a new array

Syntax

```
array.toReversed()
```

Parameters

NONE

Return Value

Type	Description
Array	A new array with the items reversed.

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Array Iteration Methods

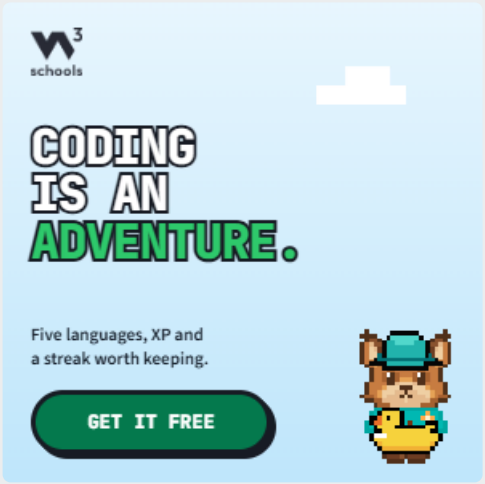
Browser Support

`toReversed()` is a JavaScript 2023 feature.

ES 2023 is supported in all modern browsers since **July 2023**:

Chrome	Edge	Firefox	Safari	Opera
110	110	115	16.4	96
Feb 2023	Feb 2023	Jul 2023	Mar 2023	May 2023

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JavaScript Array toSorted()

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Examples

```
// Create an Array
const fruits = ["Banana", "Orange", "Apple", "Mango"];

// Sort the Array
const fruit2 = fruits.toSorted();
```

Try it Yourself »

More Examples Blow !

Description

SHARE

The `toSorted()` method sorts the elements of an array in alphabetical order.

The `toSorted()` method returns a new array.

The `toSorted()` method **does not** overwrite the original array.

The `toSorted()` method is the **copying version** of the `sort()` method.

Array Sort Methods:

Method	Finds
reverse()	Reverses the order of the elements in an array
sort()	Sorts the elements of an array
toReversed()	Reverses the elements of an array into a new array
toSorted()	Sorts the elements of an array into a new array

Sort Compare Function

Sorting alphabetically works well for strings ("Apple" comes before "Banana").

But, sorting numbers can produce incorrect results.

"25" is bigger than "100", because "2" is bigger than "1".

You can fix this by providing a "compare function" (See examples below).

Syntax

```
array.sort(compareFunction)
```

Parameters

Parameter	Description
compareFunction	Optional. A function that defines a sort order. The function should return a negative, zero, or positive value, depending on the arguments: • function(a, b){return a-b} When sort() compares two values, it sends the values to the compare function, and sorts the values according to the returned (negative, zero, positive) value. **Example:** The sort function will sort 40 as a value lower than 100. When comparing 40 and 100, sort() calls the function(40,100). The function calculates 40-100, and returns -60 (a negative value).

Return Value

Type	Description
Array	A new array with the items sorted.

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More Examples

Sort Descending

Sort and then reverse the order:

```
// Create an Array
const fruits = ["Banana", "Orange", "Apple", "Mango"];

// Sort the Array
const fruits2 = fruits.toSorted();

// Reverse the Array
fruits2.reverse();
```

Try it Yourself »

Numeric Sorts

Using a Sort Function

Sort numbers in ascending order:

```
// Create an Array
const points = [40, 100, 1, 5, 25, 10];

// Sort the Array
let points2 = points.toSorted(function(a, b){return a-b});
```

Try it Yourself »

Sort numbers in descending order:

```
// Create an Array
const points = [40, 100, 1, 5, 25, 10];

// Sort the Array
let points2 = points.toSorted(function(a, b){return b-a});
```

Try it Yourself »

Find the lowest value:

```
// Create an Array
const points = [40, 100, 1, 5, 25, 10];

// Sort the numbers in ascending order
let points2 = points.toSorted(function(a, b){return a-b});

let lowest = points2[0];
```

Try it Yourself »

Find the highest value:

```
// Create an Array
const points = [40, 100, 1, 5, 25, 10];

// Sort the numbers in descending order:
let points2 = points.toSorted(function(a, b){return b-a});

let highest = points2[0];
```

Try it Yourself »

Find the highest value:

```
// Create an Array
const points = [40, 100, 1, 5, 25, 10];
```

```
// Sort the numbers in ascending order:
let points2 = points.toSorted(function(a, b){return a-b});

let highest = points2[points.length-1];
```

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Browser Support

`toSorted()` is a JavaScript 2023 feature.

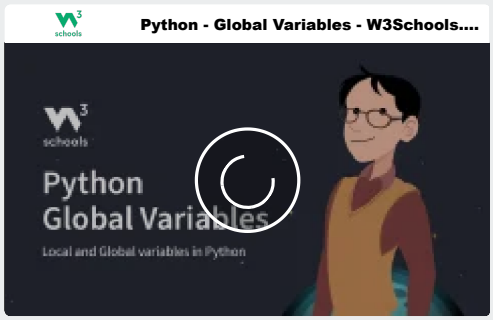
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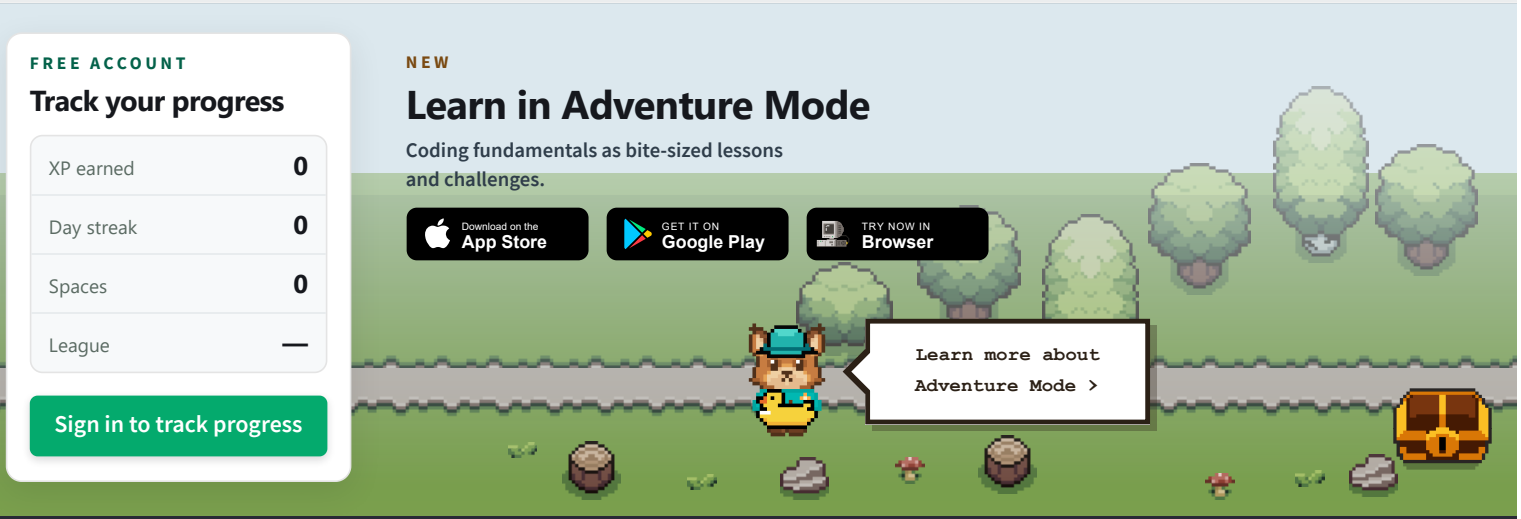
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JavaScript Array toSpliced()

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Examples

```
// Create an Array
const fruits = ["Banana", "Orange", "Apple", "Mango"];

// At position 2, add "Lemon" and "Kiwi":
const fruits2 = fruits.toSpliced(2, 0, "Lemon", "Kiwi");
```

Try it Yourself »

More Examples Below !

Description

SHARE

The `toSpliced()` method adds and/or removes array elements.

The `toSpliced()` method returns a new array.

The `toSpliced()` method **does not** change the original array.

The `toSpliced()` method is the **copying version** of the `splice()` method.

See Also:

The Array splice() Method

The Array slice() Method

Syntax

```
array.toSpliced(index, count, item1, ....., itemX)
```

Parameters

Parameter	Description
index	Required. The index (position) to add or remove items. A negative value counts from the end of the array.
count	Optional. Number of items to be removed.
item1,...	Optional. The new elements(s) to be added.

Return Value

Type	Description
Array	A new array including the changes.

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More Examples

Example

```
// Create an Array
const fruits = ["Banana", "Orange", "Apple", "Mango"];

// At position 2, remove 2 items
const fruits2 = fruits.toSpliced(2, 2);
```

Try it Yourself »

Example

```
// Create an Array
const fruits = ["Banana", "Orange", "Apple", "Mango"];

// At position 2, remove 1 item, add "Lemon" and "Kiwi"
const fruits2 = fruits.toSpliced(2, 1, "Lemon", "Kiwi");
```

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Browser Support

`toSpliced()` is a JavaScript 2023 feature.

ES 2023 is supported in all modern browsers since **July 2023**:

				
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JavaScript Array toString()

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Example

Convert an array to a string:

```
const fruits = ["Banana", "Orange", "Apple", "Mango"];
let text = fruits.toString();
```

Try it Yourself »

Description

SHARE

The `toString()` method returns a string with array values separated by commas.

The `toString()` method does not change the original array.

Note

Every JavaScript object has a `toString()` method.

The `toString()` method is used by JavaScript when an object needs to be displayed as a text (like in HTML), or when an object needs to be used as a string.

Syntax

```
array.toString()
```

Parameters

NONE

Return Value

Type	Description
A string	The array values separated by commas.

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Browser Support

`toString()` is an ECMAScript1 (JavaScript 1997) feature.

It is supported in all browsers:

				
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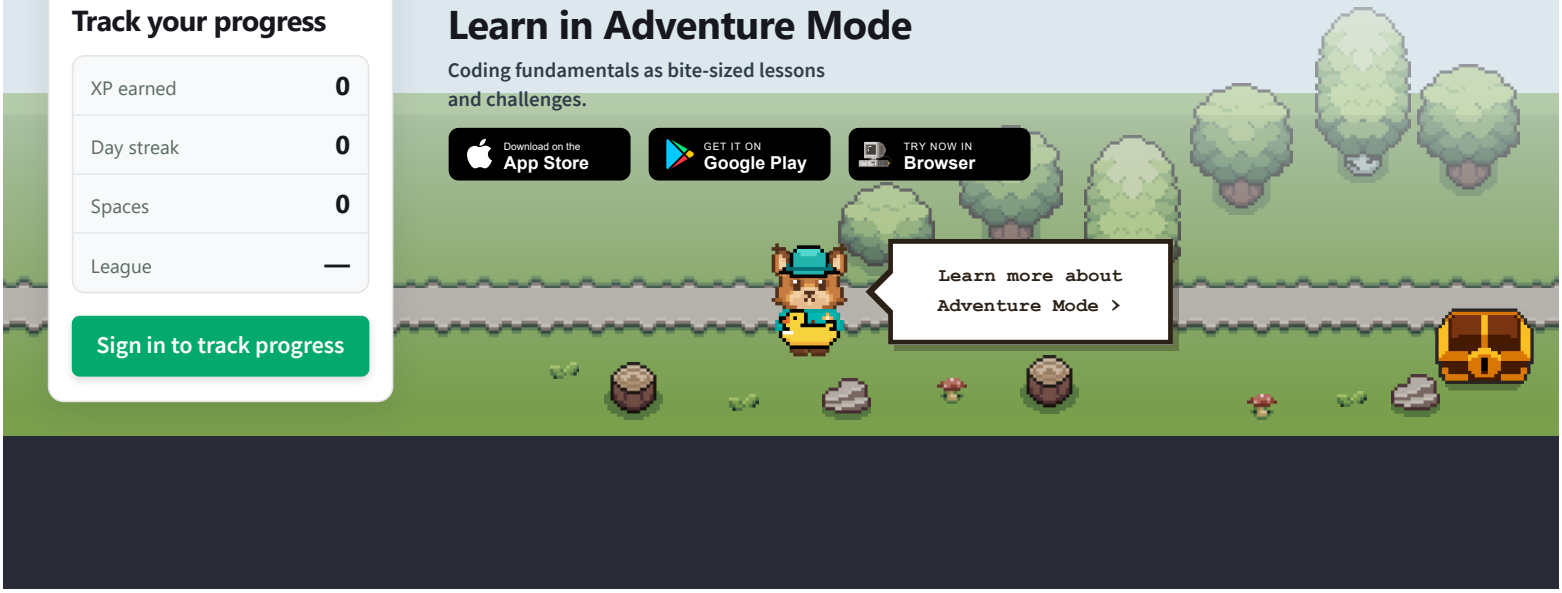
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JavaScript Array unshift()

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Example

Add new elements to an array:

```
const fruits = ["Banana", "Orange", "Apple", "Mango"];
fruits.unshift("Lemon","Pineapple");
```

Try it Yourself »

Description

SHARE

The `unshift()` method adds new elements to **the beginning** of an array.

The `unshift()` method overwrites the original array.

See Also:

The Array shift() Method

The Array push() Method

The Array pop() Method

Syntax

```
array.unshift(item1, item2, ..., itemX)
```

Parameters

Type	Description
item1 *item2* .. *itemX*	The item(s) to add to the array. Minimum one item is required.

Return Value

Type	Description
A number	The new length of the array.

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Browser Support

`unshift()` is an ECMAScript1 (JavaScript 1997) feature.

It is supported in all browsers:

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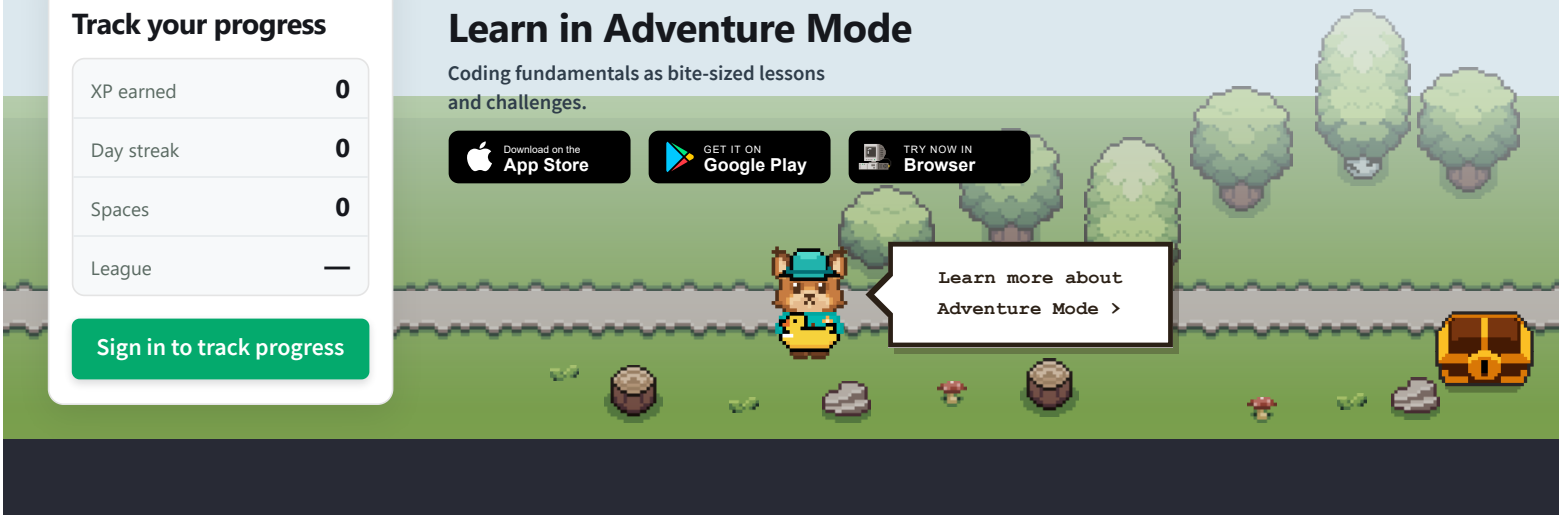
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JavaScript Array valueOf()

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Examples

Get the value of fruits:

```
const fruits = ["Banana", "Orange", "Apple", "Mango"];
const myArray = fruits.valueOf();
```

Try it Yourself »

`fruits.valueOf()` returns the same as fruits:

```
const fruits = ["Banana", "Orange", "Apple", "Mango"];
const myArray = fruits;
```

Try it Yourself »

Description

SHARE

The `valueOf()` method returns the array itself.

The `valueOf()` method does not change the original array.

`fruits.valueOf()` returns the same as `fruits`.

Syntax

```
array.valueOf()
```

Parameters

NONE

Return Value

Type	Description
An array	The `valueOf()` method returns itself!

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Browser Support

`valueOf()` is an ECMAScript1 (JavaScript 1997) feature.

It is supported in all browsers:

				
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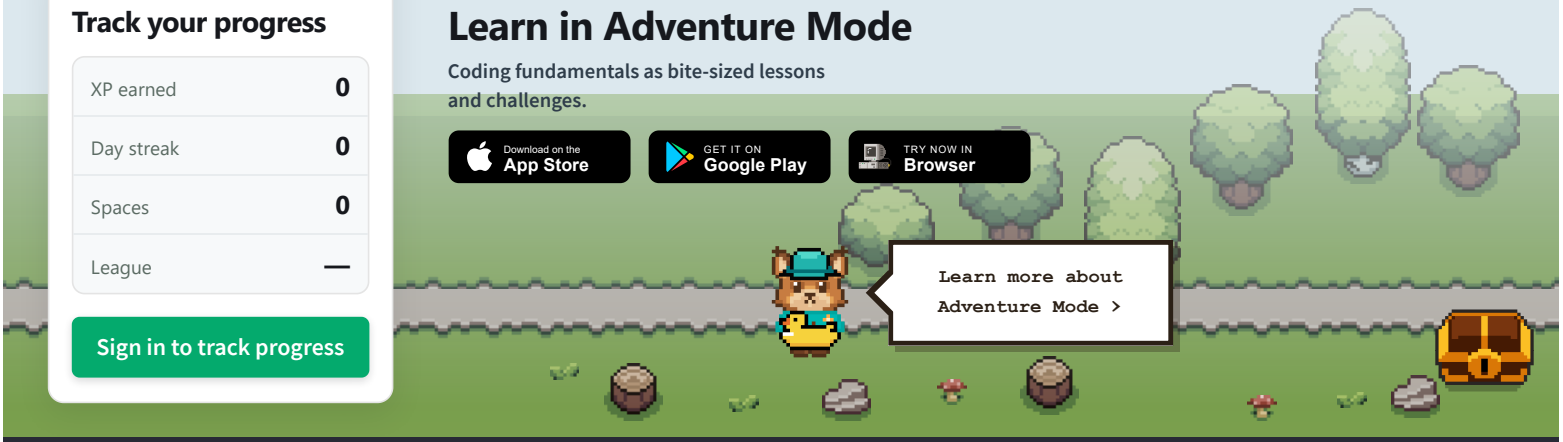
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JavaScript Array with()

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JavaScript Array with() Method

SHARE

ES2023 added the Array with() method as a safe way to update elements in an array without altering the original array.

Example

```
const months = ["Januar", "Februar", "Mar", "April"];
const myMonths = months.with(2, "March");
```

Try it Yourself »

Description

The `with()` method updates a specified array element.

The `with()` method returns a new array.

The `with()` method does not change the original array.

Syntax

```
array.with(index, value)
```

Parameters

Parameter	Description
index	Required. The index (position) of the item to change. A negative index counts from the end of the array.
value	Required. The new value.

Return Value

A new array containing the changed element.

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Browser Support

`with()` is a JavaScript 2023 feature.

ES 2023 is supported in all modern browsers since **July 2023**:

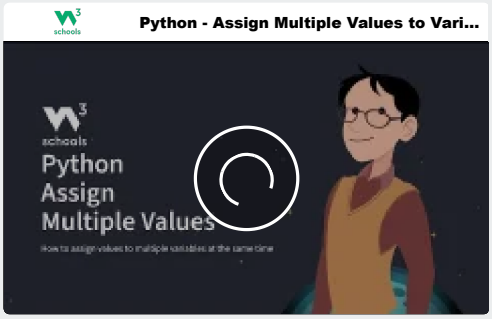
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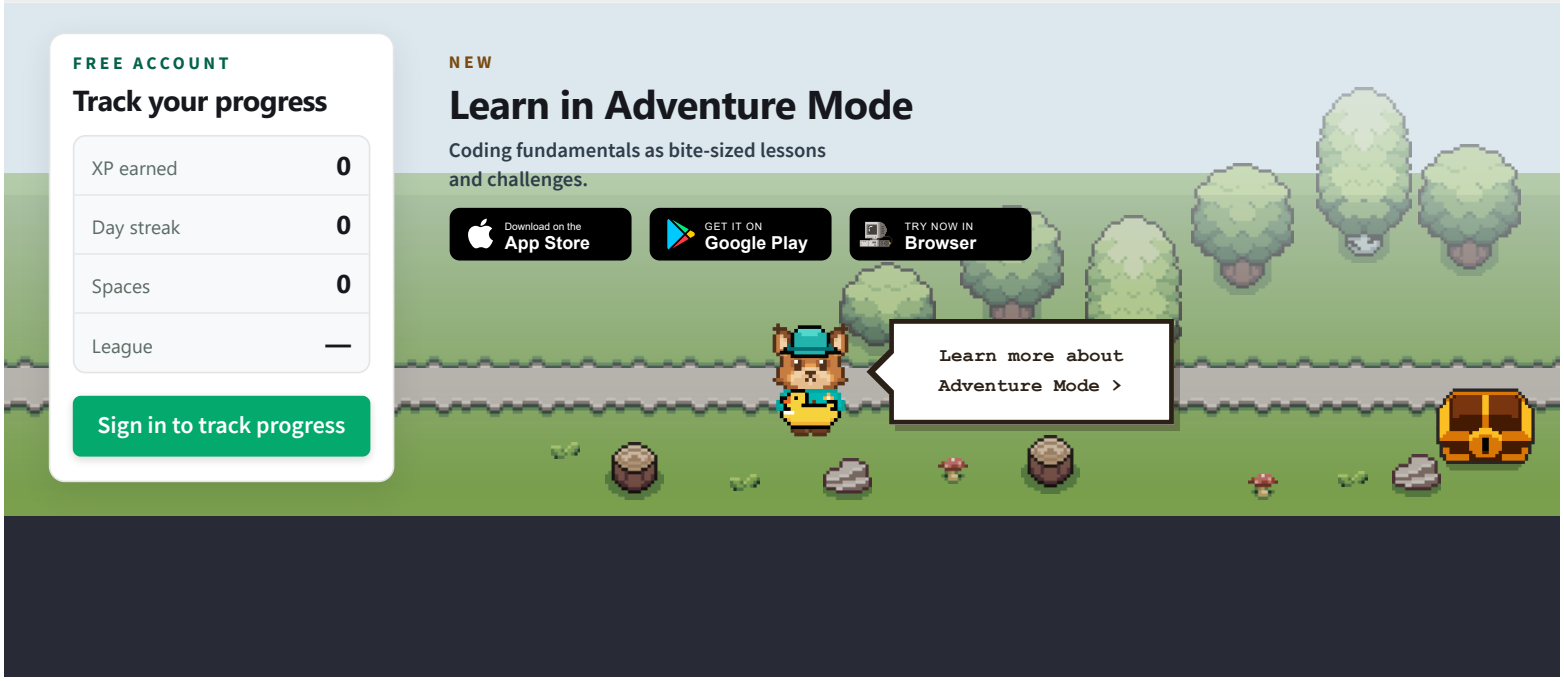
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input image | input month | input number | input password | input radio | input range | input reset | input search | input submit

input text | input time | input url | input week | kbd | label | legend | li | link | map | mark | menu | menuitem | meta

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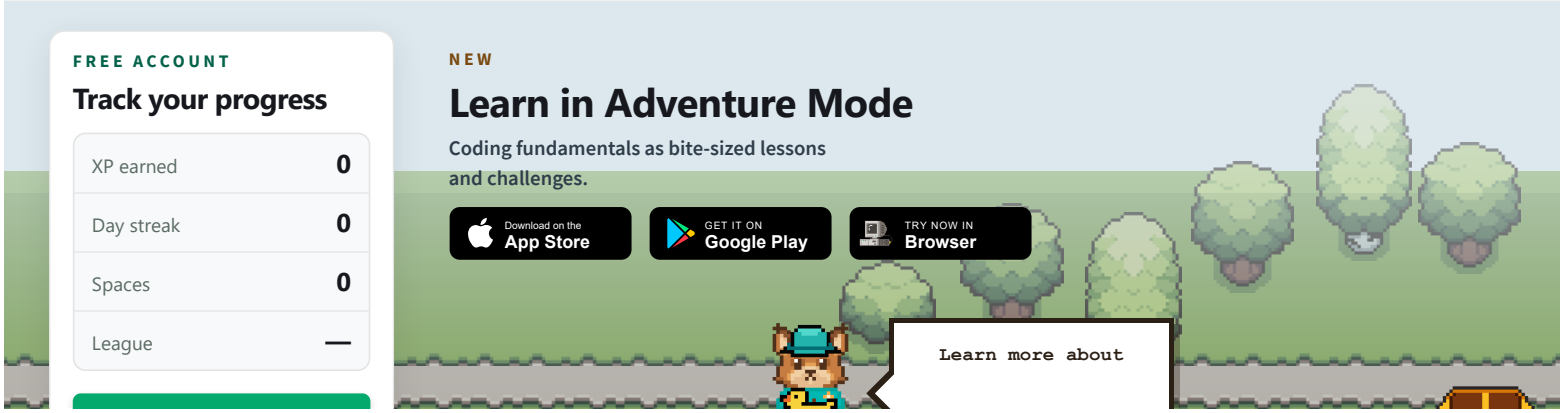
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